DEEP RESEARCH DOSSIER (OLD) • DETERMINATIVE V5.3

Sample: Daniel Ehrle (Daniel_Ehrle) | Reference: GRCh38.p14 | Modality: 48.8x BAM mean depth (Median 50.0x) WGS (panSN GBZ aligned)

This clinical genomics evidence synthesis evaluates a broad exploratory screening corpus of 1,407 prioritized candidate variants across 1,002 genes derived from a **48.8x BAM mean depth (Median 50.0x)** whole-genome sequencing (WGS) pipeline for Daniel Ehrle. To prevent cognitive bias and diagnostic over-interpretation, this report enforces rigorous domain segregation across **Monogenic Pathogenic Alleles**, **Actionable Pharmacogenomics**, **Secondary Organ-System Modifiers**, and **Endogenous Protective Factors**.

Gene	Variant	Zygosity	ClinVar	CADD (Phred)	REVEL	AlphaGenome AVI (Phred)	Associated Condition
CBLIF	`c.79+1G>A` · `rs147785187`	Heterozygous	Pathogenic	32.0	—	**33.9 (Splicing)**	CBLIF · Vitamin B12 (Cobalamin) & Folate Absorption [VCV439755]
GJB2	`p.Met34Thr` · `c.101T>C` · `rs35887622`	Heterozygous	Pathogenic	20.9	0.702	**23.6 (Cactus)**	GJB2 · Cochlear Gap Junction & Hearing Health [VCV17000]
VDR	`c.-1172A>G` · `rs4516035`	Homozygous	Likely pathogenic	5.4	—	**2.1 (DNASE-seq)**	VDR · Vitamin D Signaling & Bone Health [VCV3336650]

Gene	Diplotype	Phenotype	High-Risk Medications	Clinical Prescribing Guidance
CYP2C19	`*1/*17 (I331V (rs3758581) A>G, expression (rs12248560) C>T)`	Rapid Metabolizer	**Voriconazole, Clopidogrel, Mephenytoin, Methadone**	Increased clearance and lower trough concentrations. Perform therapeutic drug monitoring (TDM) to avoid subtherapeutic levels.
CYP2C9	`*1/*2 (R144C (rs1799853) C>T)`	Intermediate Metabolizer	**Celecoxib (Celebrex), Phenytoin (Dilantin),**	Moderate decrease in celecoxib clearance. Initiate therapy at lowest recommended dose and

Gene	Diplotype	Phenotype	High-Risk Medications	Clinical Prescribing Guidance
			Warfarin (Coumadin), Acenocoumarol	monitor for GI/cardiovascular adverse effects.
CYP3A5	`*3/*3 (splice defect (rs776746) T>C)`	Poor Metabolizer	**Carbamazepine, Clopidogrel, Cyclosporine, Efavirenz**	Genotype TT is associated with increased dose of carbamazepine in people with Epilepsy as compared to genotype CC.
F5	`rs6025 Factor V Leiden (T)/rs6025 reference (C) (p.Thr295Ala (rs371760153) [Heterozygous])`	Activated Protein C (APC) Resistance	**Combined Oral Contraceptives (Estrogen/Progestin), Hormone Replacement Therapy (Estrogen Hrt)**	Factor V Leiden carrier (rs6025 C>T heterozygous). 3- to 8-fold increased baseline venous thromboembolism (VTE) risk, amplified up to 35-fold with estrogen-containing contraceptives or HRT. Recommend progestin-only or non-hormonal contraceptives.
NAT2	`*5/*5 (I114T (rs1801280) T>C)`	Poor Metabolizer	**Isoniazid**	NAT2 *5/*5 slow acetylator phenotype. Higher risk of isoniazid-induced hepatotoxicity and peripheral neuropathy. Monitor liver function tests (LFTs) and administer pyridoxine (Vitamin B6).
SLCO1B1	`*14/*14 (N130D (rs2306283) A>G, P155T (rs11045819) C>A)`	Increased Function	**Atorvastatin (Lipitor), Rosuvastatin (Crestor), Simvastatin (Zocor)**	Standard therapeutic efficacy and low myopathy risk.
UGT1A1	`*1/*28_80 (5'UTR (rs887829) C>T, Promoter (rs3064744) insTA)`	Intermediate Metabolizer	**Belinostat, Irinotecan (Camptosar), Atazanavir (Reyataz)**	Monitor for increased hematologic toxicity.
VKORC1	`*H1/*H6 (Intron (rs2359612) G>A, Intron (rs8050894) C>G)`	Altered Enzymatic / Pathway Activity	**CPIC Substrates**	Titrate dosing per CPIC.

ANALYTICAL CALLSET CONFIDENCE

98%

High-depth 48.8x BAM mean depth (Median 50.0x) WGS callset aligned against panSN GBZ pan-genome graph; >99.8% callable SNV precision.

PRIMARY PATHOGENIC CARRIER STATUS (CBLIF, GJB2, VDR)

95%

Orthogonal concordance between ClinVar Pathogenic curation records, CADD Phred scores, and AlphaGenome AVI scores.

CLINICAL DOMAIN GROUPINGS • LANDSCAPE LAYOUT

Organized into distinct clinical categories. Formatted with page breaks and landscape orientation for clean multi-page review.

1. Pathogenic & Monogenic Carriers

4 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessions
CBLIF	c.79+1G>A	chr11:59845374	rs147785187	Heterozygous	**Tier1**	Pathogenic	32.0	—	33.9	VCV439755 /
CDKN2B	c.*2619C>T	chr9:22003368	rs1063192	Heterozygous	Tier3	Likely pathogenic|protective	0.2	—	1.3	VCV812639 /
GJB2	p.Met34Thr	chr13:20189481	rs35887622	Heterozygous	**Tier1**	Pathogenic	20.9	0.702	23.6	VCV17000 / A
VDR	c.-1172A>G	chr12:47906043	rs4516035	Homozygous	Tier3	Likely pathogenic	5.4	—	2.1	VCV3336650

2. Actionable Pharmacogenomics & Drug Response

71 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Access
ABCB1	c.*193A>G	chr7:87504050	rs3842	Heterozygous	Tier3	drug response	0.2	—	1.1	VCV82
ABCB1	c.*89A>T	chr7:87504154	rs17064	Heterozygous	Tier3	drug response	2.3	—	8.7	VCV82
ABCB1	c.2065-76T>A	chr7:87544351	rs1922242	Heterozygous	Tier3	drug response	0.7	—	0.8	VCV82
ABCB1	c.2064+73A>G	chr7:87544750	rs2235046	Heterozygous	Tier3	drug response	0.5	—	5.8	VCV82
ABCB1	c.1888-75C>T	chr7:87545074	rs28381920	Heterozygous	Tier3	drug response	2.3	—	3.7	VCV82
ABCB1	c.1350+44C>T	chr7:87550127	rs2032588	Heterozygous	Tier3	drug response	0.7	—	0.5	VCV82
ABCB1	c.1000-44G>A	chr7:87550882	rs10276036	Heterozygous	Tier3	drug response	0.2	—	0.6	VCV82
ABCB1	c.827+649A>C	chr7:87560614	rs868755	Heterozygous	Tier3	drug response	1.0	—	0.1	VCV82
ABCB1	c.702+280G>A	chr7:87565790	rs1202170	Homozygous	Tier3	drug response	0.2	—	0.1	VCV82
ABCB1	c.530+213A>G	chr7:87566572	rs28381848	Heterozygous	Tier3	drug response	8.6	—	13.0	VCV82
ABCB1	c.530+139C>T	chr7:87566646	rs1202168	Heterozygous	Tier3	drug response	3.1	—	6.4	VCV82
ABCB1	c.338+306C>T	chr7:87569866	rs1016793	Heterozygous	Tier3	drug response	0.2	—	1.5	VCV82
ABCB1	c.338+224G>T	chr7:87569948	rs2235021	Heterozygous	Tier3	drug response	0.7	—	3.1	VCV82
ABCB1	c.338+223A>T	chr7:87569949	rs2235020	Heterozygous	Tier3	drug response	1.2	—	1.8	VCV82
ABCB1	c.118-118T>G	chr7:87585798	rs1211152	Homozygous	Tier3	drug response	0.3	—	0.7	VCV82
ABCB1	c.117+502A>G	chr7:87595264	rs28381817	Heterozygous	Tier3	drug response	2.0	—	0.7	VCV82
ABCB1	c.-1A>G	chr7:87600185	rs2214102	Homozygous	Tier3	drug response	7.0	—	5.1	VCV82
ACP1	c.231+53C>T	chr2:272203	rs11553746	Heterozygous	Tier3	drug response	14.7	0.143	24.3	VCV39
ADRB2	p.Gly16Arg	chr5:148826877	rs1042713	Heterozygous	Tier3	drug response	17.9	0.079	19.5	VCV22
APOE	c.-286T>G	chr19:44905579	rs405509	Heterozygous	Tier3	drug response	10.2	—	5.2	VCV61

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Access
APOE	p.Arg176Cys	chr19:44908822	rs7412	Heterozygous	Tier2	drug response	26.0	0.588	23.4	VCV178
APOE	c.*1944T>C	chr19:44911194	rs439401	Homozygous	Tier3	drug response	5.6	—	7.9	VCV619
ATIC	c.1503+675T>C	chr2:215347616	rs4673993	Heterozygous	Tier3	drug response	2.1	—	0.8	VCV22
CERT1	c.1670-3702A>G	chr5:75360714	rs12916	Heterozygous	Tier3	drug response	3.8	—	5.7	VCV180
CHI3L1	p.Arg145Gly	chr1:203183673	rs880633	Homozygous	Tier3	drug response	23.4	0.098	18.5	VCV39
COMT	c.*522G>A	chr22:19969258	rs165599	Heterozygous	Tier3	drug response	1.8	—	4.5	VCV82
COMT	c.*764C>T	chr22:19969500	rs165728	Homozygous	Tier3	drug response	1.0	—	0.4	VCV82
CYP19A1	c.145+418G>T	chr15:51242350	rs727479	Heterozygous	Tier3	drug response	0.0	—	1.1	VCV25
CYP2C19	c.-806C>T	chr10:94761900	rs12248560	Heterozygous	Tier3	drug response	0.6	—	0.5	VCV39
CYP2C9	p.Arg144Cys	chr10:94942290	rs1799853	Heterozygous	Tier2	Likely benign|drug response|other	17.6	0.340	10.1	VCV84
CYP3A5	c.219-237A>G	chr7:99672916	rs776746	Homozygous	Tier3	association|drug response|risk factor	3.4	—	20.6	VCV50
DIO1	c.682-34C>A	chr1:53909897	rs2235544	Homozygous	Tier3	drug response	2.3	—	3.9	VCV25
DPYD	p.Val732Ile	chr1:97305364	rs1801160	Heterozygous	Tier2	drug response	23.0	0.297	21.0	VCV100
DPYD	p.Met166Val	chr1:97699535	rs2297595	Homozygous	Tier2	drug response	25.6	0.523	22.4	VCV100
DPYD	p.Cys29Arg	chr1:97883329	rs1801265	Heterozygous	Tier2	drug response	18.9	—	14.7	VCV43
EPAS1	c.1035-7C>G	chr2:46376532	rs7557402	Homozygous	Tier3	drug response	7.6	—	13.6	VCV33
F5	p.Arg534Gln	chr1:169549811	rs6025	Heterozygous	Tier2	drug response	27.9	—	23.6	VCV64
HSD3B1	p.Thr367Asn	chr1:119514623	rs1047303	Heterozygous	Tier3	Benign|drug response	0.0	0.126	7.8	VCV127
MTHFR	p.Ala222Val	chr1:11796321	rs1801133	Homozygous	Tier2	drug response	27.3	0.842	25.8	VCV35

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Access
NAT2	p.Ile114Thr	chr8:18400344	rs1801280	Homozygous	Tier2	Benign|drug response	18.0	0.098	11.5	VCV72
NPC1L1	c.2637+1287A>G	chr7:44530468	rs11763759	Heterozygous	Tier3	drug response	8.9	—	1.0	VCV18
NPNT	p.Gln159His	chr4:105938392	rs35132891	Homozygous	Tier3	drug response	22.3	0.431	20.9	VCV39
OPRM1	c.291-2259G>A	chr6:154087567	rs9322446	Homozygous	Tier3	drug response	0.2	—	0.5	VCV82
OPRM1	c.644-83G>C	chr6:154090869	rs2075572	Homozygous	Tier3	drug response	0.1	—	0.6	VCV82
OPRM1	c.1164+1768A>C	chr6:154093240	rs9282821	Homozygous	Tier3	drug response	7.5	—	0.3	VCV82
OPRM1	c.1164+1839A>T	chr6:154093311	rs540825	Homozygous	Tier3	drug response	1.7	0.095	2.1	VCV82
OPRM1	c.1164+1956A>G	chr6:154093428	rs675026	Homozygous	Tier3	drug response	2.0	—	0.1	VCV82
OPRM1	c.1164+1966C>T	chr6:154093438	rs562859	Homozygous	Tier3	drug response	10.8	—	10.2	VCV82
OPRM1	c.1165-11116A>G	chr6:154107567	rs650245	Homozygous	Tier3	drug response	3.0	—	1.2	VCV82
OPRM1	c.1165-10726del	chr6:154107951	chr6:154107951	Homozygous	Tier3	drug response	—	—	—	VCV82
OPRM1	c.1165-8425C>G	chr6:154110258	rs497332	Homozygous	Tier3	drug response	3.4	—	4.9	VCV82
SCN1A	c.603-91G>A	chr2:166053034	rs3812718	Homozygous	Tier2	drug response	20.4	—	23.1	VCV12
SLC19A1	p.His27Arg	chr21:45537880	rs1051266	Homozygous	Tier2	drug response	7.5	0.192	9.0	VCV15
SLC29A3	p.Ser158Phe	chr10:71351651	rs780668	Homozygous	Tier2	Benign|drug response	23.5	0.286	23.6	VCV13
UGT1A4	c.868-9898G>A	chr2:233757136	rs10929302	Heterozygous	Tier3	drug response	5.2	—	5.2	VCV81
UGT1A4	c.868-6787_868-6786dup	chr2:233760234	chr2:233760234	Heterozygous	Tier2	Conflicting classifications of pathogenicity|drug response|other	—	—	—	VCV12

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Access
UGT2B7	c.-161T>C	chr4:69096360	rs7668258	Heterozygous	Tier3	drug response	0.3	—	0.5	VCV82
UGT2B7	c.721+201A>C	chr4:69097442	rs12647681	Heterozygous	Tier3	drug response	0.6	—	8.2	VCV82
UGT2B7	c.721+209A>G	chr4:69097450	rs12647682	Heterozygous	Tier3	drug response	0.2	—	10.0	VCV82
UGT2B7	c.721+272T>C	chr4:69097513	rs12648462	Heterozygous	Tier3	drug response	4.7	—	5.3	VCV82
UGT2B7	c.722-78T>C	chr4:69098462	rs7439326	Heterozygous	Tier3	drug response	1.8	—	4.5	VCV82
UGT2B7	c.722-49A>G	chr4:69098491	rs7438244	Heterozygous	Tier3	drug response	0.5	—	6.7	VCV82
UGT2B7	p.Tyr268His	chr4:69098620	rs7439366	Heterozygous	Tier3	drug response	6.5	0.066	10.9	VCV82
UGT2B7	p.Leu353=	chr4:69107231	rs4292394	Heterozygous	Tier3	drug response	0.1	—	2.3	VCV82
UGT2B7	c.1090+64A>T	chr4:69107326	rs4337789	Heterozygous	Tier3	drug response	0.1	—	2.6	VCV82
UGT2B7	c.*296T>C	chr4:69113032	rs6851533	Heterozygous	Tier3	drug response	6.7	—	4.9	VCV82
VKORC1	c.283+837T>C	chr16:31092475	rs2359612	Heterozygous	Tier2	drug response	0.5	—	12.6	VCV22
VKORC1	c.283+124G>C	chr16:31093188	rs8050894	Heterozygous	Tier2	drug response	0.7	—	5.4	VCV22
VKORC1	c.174-136C>T	chr16:31093557	rs9934438	Heterozygous	Tier2	drug response	14.2	—	22.7	VCV37
VKORC1	c.-1639G>A	chr16:31096368	rs9923231	Heterozygous	Tier2	drug response	1.6	—	2.6	VCV22
XRCC1	p.Gln399Arg	chr19:43551574	rs25487	Heterozygous	Tier3	drug response	14.4	0.072	8.8	VCV22

3. Cardiovascular & Channelopathy Modifiers

183 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
ABCG5	p.Phe624Leu	chr2:43813202	rs150401285	Heterozygous	Tier2	Uncertain significance	24.1	0.127	26.9	VCV
ABCG5	p.Gly27Ala	chr2:43838600	rs56204478	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.4	0.244	15.3	VCV
ADNP	c.-691_-690dup	chr20:50931252	chr20:50931252	Heterozygous	Tier2	Not reviewed	—	—	—	— /
ALMS1	p.Ser2101Leu	chr2:73452829	rs28730854	Heterozygous	Tier3	Benign	15.3	—	16.9	VCV
ALMS1	c.9907+129T>G	chr2:73535078	rs73947814	Heterozygous	Tier2	Likely benign	6.0	—	9.5	VCV
APC2	p.Arg1666Pro	chr19:1468298	rs767618250	Heterozygous	Tier2	Uncertain significance	7.9	0.398	0.9	VCV
APOB	p.Ala4481Thr	chr2:21001981	rs1801695	Heterozygous	Tier2	Conflicting classifications of pathogenicity	7.7	0.033	5.1	VCV
APOB	p.Ser4338Asn	chr2:21002409	rs1042034	Heterozygous	Tier2	Benign	0.8	0.055	4.3	VCV
APOB	p.Pro2739Leu	chr2:21008652	rs676210	Heterozygous	Tier2	Benign/Likely benign	28.0	0.373	23.9	VCV
APOB	p.Thr2515=	chr2:21009323	rs693	Heterozygous	Tier2	Benign	5.4	—	10.5	VCV
APOB	p.Ile2313Val	chr2:21009931	rs584542	Homozygous	Tier2	Benign	0.1	0.014	3.6	VCV
APOB	p.Asp2312=	chr2:21009932	rs1041968	Heterozygous	Tier2	Benign	0.5	—	7.0	VCV
APOB	c.3843-77T>C	chr2:21013610	rs488329	Homozygous	Tier2	Benign	6.3	—	3.1	VCV
APOB	p.Ala618Val	chr2:21028042	rs679899	Heterozygous	Tier2	Benign/Likely benign	24.8	0.429	22.8	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
APOB	c.693+360T>C	chr2:21036740	rs520354	Heterozygous	Tier2	Benign	4.1	—	1.8	VCV
APOB	c.237+92G>T	chr2:21042269	rs661665	Homozygous	Tier2	Benign	2.3	—	12.3	VCV
B9D1	p.Thr190Ser	chr17:19343366	rs147684440	Heterozygous	Tier2	Conflicting classifications of pathogenicity	12.2	0.047	16.5	VCV
BCHE	p.Phe146ValfsTer12	chr3:165830599	rs774072493	Heterozygous	Tier2	Conflicting classifications of pathogenicity	6.7	—	—	VCV
BCHE	p.Gly145=	chr3:165830599	rs71534242	Heterozygous	Tier2	Conflicting classifications of pathogenicity	6.4	—	10.4	VCV
BRD4	p.Arg1097His	chr19:15239814	rs35676845	Heterozygous	Tier3	Benign/Likely benign	24.2	0.091	22.3	VCV
CACNA1C	p.Gly37Arg	chr12:2115283	rs34534613	Heterozygous	Tier2	Benign/Likely benign	26.0	0.655	29.1	VCV
CACNA1C	c.2104-120del	chr12:2582700	chr12:2582700	Heterozygous	Tier2	Not reviewed	—	—	—	— /
CACNA1S	p.Leu1800Ser	chr1:201040054	rs12139527	Heterozygous	Tier2	Benign	24.9	0.205	23.6	VCV
CACNA1S	p.Arg1539Cys	chr1:201047168	rs3850625	Heterozygous	Tier2	Benign	26.8	0.281	23.6	VCV
CACNA1S	c.4543+7G>A	chr1:201047518	rs2297905	Heterozygous	Tier2	Benign	3.1	—	1.4	VCV
CACNA1S	c.4113+7T>C	chr1:201050977	rs6702590	Homozygous	Tier2	Benign	3.1	—	0.1	VCV
CACNA1S	p.Phe801=	chr1:201069559	rs7415038	Heterozygous	Tier2	Benign	12.2	—	16.2	VCV
CACNA1S	c.1828-5T>C	chr1:201075620	rs1998721	Homozygous	Tier2	Benign/Likely benign	0.1	—	4.1	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
CACNA1S	p.Gly517=	chr1:201077947	rs4915477	Heterozygous	Tier2	Benign	3.4	—	0.0	VCV
CACNA1S	p.Cys505=	chr1:201077983	rs9427714	Heterozygous	Tier2	Benign	12.6	—	18.1	VCV
CACNA1S	p.Leu458His	chr1:201083182	rs12742169	Heterozygous	Tier2	Benign	23.4	0.302	8.7	VCV
CACNA1S	p.Pro457=	chr1:201083184	rs12742170	Heterozygous	Tier2	Benign	4.5	—	0.0	VCV
CACNA1S	p.Ile199=	chr1:201091737	rs2296383	Homozygous	Tier2	Benign	8.1	—	1.2	VCV
CACNA1S	c.-231A>G	chr1:201112570	rs4506487	Heterozygous	Tier2	Benign	8.8	—	9.7	VCV
CDKN1C	p.Ala202_Pro205del	chr11:2884861	chr11:2884861	Heterozygous	Tier3	Benign/Likely benign	—	—	—	VCV
CEP41	c.*1056A>C	chr7:130397835	rs574982700	Heterozygous	Tier2	Uncertain significance	14.9	—	17.4	VCV
COL2A1	p.Glu142Asp	chr12:47997874	rs34392760	Heterozygous	Tier3	Benign/Likely benign	21.1	0.289	15.6	VCV
COL6A1	p.Ser116Asn	chr21:45984388	rs11553519	Heterozygous	Tier3	Benign	10.5	0.191	15.6	VCV
DCC	c.91+22042A>G	chr18:52362920	rs8095741	Heterozygous	Tier2	Not reviewed	4.1	—	2.7	— /
DHRS3	p.Arg56Cys	chr1:12617183	rs148358973	Heterozygous	Tier2	Likely benign	28.1	—	22.3	VCV
DMD	c.94-60270G>C	chrX:32910090	rs1923093	Homozygous	Tier2	Not reviewed	1.1	—	5.4	— /
DMD	c.32-76315T>A	chrX:33096515	rs4829276	Homozygous	Tier2	Not reviewed	1.0	—	0.6	— /
DNAAF11	c.1140+8154A>G	chr8:132602012	rs4520144	Heterozygous	Tier2	Not reviewed	1.8	—	1.8	— /
DNAH7	p.Lys228=	chr2:196024488	rs555558470	Heterozygous	**Tier1**	Not reviewed	12.9	—	23.0	— /
DSC2	p.Arg798Gln	chr18:31069009	rs61731921	Heterozygous	Tier3	Benign	0.1	0.167	5.4	VCV
DUSP6	p.Thr87Ile	chr12:89351780	rs61734372	Heterozygous	Tier3	Benign	22.7	0.063	21.6	VCV
DYRK1A	c.10+25013T>C	chr21:37445397	rs9974713	Heterozygous	Tier2	Not reviewed	3.1	—	6.9	— /

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
EBP	c.301+446A>T	chrX:48524518	rs62602501	Homozygous	Tier2	Not reviewed	0.7	—	0.0	— /
ELN	p.Gly711Asp	chr7:74068657	rs41511151	Heterozygous	Tier2	Benign/Likely benign	23.0	0.340	24.2	VCV
FBLN5	c.*647C>T	chr14:91869577	rs10197	Heterozygous	Tier3	Benign/Likely benign	15.8	—	14.3	VCV
FIG4	c.2546+2077T>G	chr6:109798928	rs4330563	Heterozygous	Tier2	Not reviewed	5.1	—	3.9	— /
FIG4	c.2547-9862A>C	chr6:109815226	rs4388322	Heterozygous	Tier2	Not reviewed	0.7	—	2.2	— /
FIG4	c.2547-38009T>C	chr6:109839813	rs9400325	Heterozygous	Tier2	Not reviewed	0.7	—	0.5	— /
FIG4	c.2547-18770T>C	chr6:109859052	rs9400330	Heterozygous	Tier2	Not reviewed	2.3	—	4.4	— /
FIG4	c.2547-16414T>G	chr6:109861408	rs9481016	Heterozygous	Tier2	Not reviewed	3.2	—	5.2	— /
FLI1	c.385+966G>A	chr11:128769238	rs1306595597	Heterozygous	Tier2	Not reviewed	14.6	—	14.6	— /
FLNB	c.6773-225A>C	chr3:58155735	rs1200505096	Heterozygous	Tier2	Not reviewed	4.9	—	7.9	— /
FLVCR1	c.*2061_*2080dup	chr1:212897350	chr1:212897350	Heterozygous	Tier2	Uncertain significance	—	—	—	VCV
FOXE3	p.Glu36Lys	chr1:47416421	rs900562270	Heterozygous	Tier2	Uncertain significance	13.7	0.089	21.0	VCV
FOXP1	c.-11-12518G>A	chr3:71210910	rs1158614120	Heterozygous	Tier2	Not reviewed	12.0	—	11.9	— /
FRAS1	c.2423-391C>T	chr4:78363122	rs2100484	Heterozygous	Tier2	Not reviewed	2.1	—	1.8	— /
FRAS1	c.2722+2111G>A	chr4:78366165	rs11936250	Heterozygous	Tier2	Not reviewed	0.4	—	2.6	— /
FRAS1	c.11298+469G>A	chr4:78537669	rs7664814	Heterozygous	Tier2	Not reviewed	0.4	—	0.8	— /
GDF6	p.Ala249Glu	chr8:96145185	rs121909352	Heterozygous	Tier2	Conflicting classifications of pathogenicity	9.6	0.269	8.7	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
GJB3	p.Arg75Cys	chr1:34784985	rs370476720	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	26.2	0.943	32.4	VCV
GJB4	p.Cys169Trp	chr1:34761761	rs79193415	Heterozygous	Tier3	Benign	24.9	0.688	23.8	VCV
GLI2	c.644-5907_644-5906del	chr2:120962807	chr2:120962807	Heterozygous	Tier2	Not reviewed	—	—	—	— /
GPC6	c.877+107375del	chr13:94135255	chr13:94135255	Heterozygous	Tier2	Not reviewed	—	—	—	— /
GTPBP3	p.Arg368His	chr19:17341172	rs3745193	Heterozygous	Tier3	Benign	22.5	0.030	17.9	VCV
HIC1	p.Ala457Thr	chr17:2058059	rs201599490	Heterozygous	Tier2	Likely benign	21.7	0.082	18.7	VCV
HYDIN	p.Thr2664Arg	chr16:70918224	rs780236458	Heterozygous	Tier2	Not reviewed	1.7	0.061	6.8	— /
IKBKG	c.1056-6C>T	chrX:154563953	rs5945206	Homozygous	Tier2	Uncertain significance	10.5	—	7.3	VCV
INTS1	p.Gln1448Arg	chr7:1478872	rs185292877	Heterozygous	Tier2	Uncertain significance	22.8	0.389	22.7	VCV
JKAMP	c.717+1157A>G	chr14:59502424	rs113112586	Heterozygous	**Tier1**	Not reviewed	13.9	—	13.5	— /
KIAA0586	p.Glu1185Lys	chr14:58488646	rs200722491	Heterozygous	Tier2	Conflicting classifications of pathogenicity	31.0	0.358	24.9	VCV
KMT2C	c.4661-778C>A	chr7:152188625	rs13242182	Homozygous	Tier2	Not reviewed	1.1	—	0.2	— /
LAMA3	p.Glu1354Lys	chr18:23847592	rs34799994	Heterozygous	Tier2	Benign/Likely benign	20.9	0.061	16.2	VCV
LIG4	p.Thr9Ile	chr13:108211243	rs1805388	Heterozygous	Tier2	Benign/Likely benign	22.5	0.073	19.0	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
LMOD3	p.Glu142Asp	chr3:69119929	rs111848977	Heterozygous	Tier2	Benign/Likely benign	13.8	0.120	7.6	VCV
LRP5	p.Val233Glu	chr11:68363758	rs1177484229	Heterozygous	**Tier1**	Uncertain significance	24.9	0.903	25.4	VCV
LRRC7	p.Gln1fs	chr1:69669997	rs201453318	Heterozygous	Tier2	Not reviewed	—	—	—	— /
MAP3K20	c.1359+3075G>A	chr2:173242571	rs4972539	Heterozygous	Tier2	Not reviewed	7.7	—	11.9	— /
MLYCD	p.Arg11Gly	chr16:83899175	rs138107830	Heterozygous	Tier3	Benign	14.9	0.254	7.0	VCV
MOGS	p.Val673Ile	chr2:74461772	rs114933392	Heterozygous	Tier3	Benign/Likely benign	8.0	0.043	11.1	VCV
MRPS22	c.-71-65596T>G	chr3:139278360	rs78362422	Heterozygous	Tier2	Not reviewed	2.5	—	1.3	— /
MTTP	p.Gly661Ser	chr4:99611445	rs113337987	Heterozygous	Tier3	Benign/Likely benign	17.5	0.105	18.2	VCV
MYL2	c.-52-2913T>C	chr12:110923494	rs12828640	Homozygous	Tier2	Not reviewed	5.4	—	4.6	— /
MYL2	c.-52-15019A>G	chr12:110935600	rs4426183	Homozygous	Tier2	Not reviewed	0.5	—	0.3	— /
MYO18B	c.6157-7087G>T	chr22:25985276	rs9306420	Homozygous	Tier2	Not reviewed	13.1	—	1.5	— /
NDUFS3	p.Ser45fs	chr11:47579999	chr11:47579999	Heterozygous	Tier2	Not reviewed	—	—	—	— /
NEB	p.Arg2773Gln	chr2:151642629	rs35974308	Heterozygous	Tier3	Benign	22.8	0.272	23.5	VCV
NEB	p.Glu1469Asp	chr2:151671122	rs34800215	Heterozygous	Tier3	Benign	21.5	0.093	20.1	VCV
NEB	p.Thr1424=	chr2:151672396	rs35654397	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.1	—	0.2	VCV
NEB	c.3147+5G>A	chr2:151679913	rs74859201	Heterozygous	Tier2	Benign	15.5	—	17.3	VCV
NEK10	c.2971-433A>G	chr3:27132423	rs1267845859	Heterozygous	Tier2	Not reviewed	0.8	—	14.6	— /

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
NEK10	c.362+248G>A	chr3:27344024	rs113156482	Heterozygous	**Tier1**	Not reviewed	12.8	—	20.4	— /
ODAD1	p.Ser676Asn	chr19:48297073	rs74351635	Heterozygous	Tier3	Benign	22.2	0.139	20.2	VCV
PAX3	p.Thr315Lys	chr2:222221236	rs2234675	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.8	0.396	25.8	VCV
PCSK9	c.207+15A>G	chr1:55040059	rs2495482	Homozygous	Tier2	Benign	8.7	—	3.8	VCV
PCSK9	p.Gln342=	chr1:55057360	rs509504	Homozygous	Tier2	Conflicting classifications of pathogenicity	8.6	—	13.3	VCV
PCSK9	p.Val460=	chr1:55058524	rs540796	Homozygous	Tier2	Benign	8.8	—	15.2	VCV
PCSK9	p.Val474Ile	chr1:55058564	rs562556	Homozygous	Tier2	Benign/Likely benign	0.0	0.051	0.2	VCV
PCSK9	p.Gly670Glu	chr1:55063514	rs505151	Homozygous	Tier2	Benign/Likely benign	1.7	0.067	12.1	VCV
PCSK9	c.*571C>T	chr1:55064155	rs662145	Homozygous	Tier2	Benign	0.7	—	0.3	VCV
PEX6	p.Val788Met	chr6:42966044	rs267608240	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	35.0	0.921	32.3	VCV
PIEZO2	p.Thr1834=	chr18:10708361	rs543259433	Heterozygous	Tier3	Likely benign	16.1	—	13.5	VCV
PKP2	p.Glu58Asp	chr12:32896558	rs146708884	Heterozygous	Tier2	Benign/Likely benign	23.1	0.175	23.0	VCV
PLCH1	p.Arg1406His	chr3:155481809	rs150381264	Heterozygous	Tier2	Likely benign	0.2	0.073	5.3	VCV
PLEC	p.Thr4402Met	chr8:143916616	rs113513807	Heterozygous	Tier3	Benign	5.2	0.252	4.3	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
PLEC	p.Gly3353Ala	chr8:143919763	rs35261863	Heterozygous	Tier3	Benign	23.2	0.339	23.9	VCV
PLEC	p.Thr3186=	chr8:143920263	rs201102719	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.0	—	0.0	VCV
PNPLA6	c.2261-1027C>G	chr19:7552848	rs114424868	Heterozygous	Tier2	Not reviewed	3.7	—	4.6	— /
POGZ	c.284-4376T>C	chr1:151435217	rs11578888	Homozygous	Tier2	Not reviewed	0.1	—	0.5	— /
PTPN11	c.15-604G>A	chr12:112445672	rs11066309	Heterozygous	Tier2	Not reviewed	0.3	—	0.5	— /
RAB3GAP2	c.*1821C>T	chr1:220149430	rs144988111	Heterozygous	Tier2	Uncertain significance	6.8	—	14.9	VCV
RAD51	c.-3+203G>T	chr15:40695628	rs5030791	Heterozygous	**Tier1**	Not reviewed	5.5	—	19.1	VCV
RBM20	c.192-37897A>C	chr10:110742904	rs869432	Heterozygous	Tier2	Not reviewed	0.4	—	0.0	— /
RBM20	c.192-27273T>C	chr10:110753528	rs2840311	Heterozygous	Tier2	Not reviewed	6.1	—	3.5	— /
RECQL4	p.Arg754Trp	chr8:144513421	rs193024188	Heterozygous	Tier2	Conflicting classifications of pathogenicity	22.1	—	17.7	VCV
ROBO4	c.1319-30C>T	chr11:124894075	rs1051723367	Heterozygous	Tier2	Not reviewed	2.9	—	16.7	— /
RTL1	p.Arg1215His	chr14:100881145	rs61993318	Heterozygous	Tier2	Benign/Likely benign	0.7	0.024	0.1	VCV
RTL1	p.Arg1108Gln	chr14:100881466	rs117476938	Heterozygous	Tier3	Benign	3.5	0.015	0.7	VCV
RYR1	p.Leu198=	chr19:38444640	rs2229139	Heterozygous	Tier2	Benign/Likely benign	10.4	—	12.1	VCV
RYR1	p.Ala359=	chr19:38448768	rs10406027	Homozygous	Tier2	Benign	0.4	—	0.0	VCV
RYR1	p.Ser556=	chr19:38455542	rs2288888	Heterozygous	Tier2	Benign	0.7	—	0.0	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
RYR1	p.Pro762=	chr19:38459264	rs3745847	Heterozygous	Tier2	Benign	0.2	—	0.0	VCV
RYR1	p.Asn993=	chr19:38466199	rs2228070	Homozygous	Tier2	Benign	0.9	—	0.0	VCV
RYR1	p.Ile1152=	chr19:38469040	rs11083462	Homozygous	Tier2	Benign	5.6	—	0.0	VCV
RYR1	p.Glu1874=	chr19:38489251	rs35021937	Heterozygous	Tier2	Benign/Likely benign	5.4	—	3.6	VCV
RYR1	p.His2420=	chr19:38499953	rs12973632	Heterozygous	Tier2	Benign	2.3	—	0.0	VCV
RYR1	c.7614+10C>G	chr19:38501000	rs2960323	Heterozygous	Tier2	Benign	4.9	—	6.2	VCV
RYR1	c.7835+5A>G	chr19:38502732	rs1469695	Heterozygous	Tier2	Benign	2.2	—	9.1	VCV
RYR1	p.Arg2624=	chr19:38502916	rs1469698	Heterozygous	Tier2	Benign	7.1	—	15.8	VCV
RYR1	p.Thr2659=	chr19:38504270	rs2229144	Heterozygous	Tier2	Benign	0.5	—	0.0	VCV
RYR1	p.Ile2706=	chr19:38504798	rs2960340	Heterozygous	Tier2	Benign	0.8	—	0.2	VCV
RYR1	p.Asp2730=	chr19:38504870	rs2915951	Heterozygous	Tier2	Benign	0.3	—	0.0	VCV
RYR1	p.Glu2779=	chr19:38505335	rs2915952	Heterozygous	Tier2	Benign	6.4	—	12.9	VCV
RYR1	p.Ser2863=	chr19:38506350	rs2229146	Heterozygous	Tier2	Benign	3.7	—	0.1	VCV
RYR1	c.8693-10G>C	chr19:38506819	rs2915958	Heterozygous	Tier2	Benign	5.2	—	4.1	VCV
RYR1	p.Pro3062=	chr19:38512085	rs2071089	Heterozygous	Tier2	Benign	0.1	—	0.0	VCV
RYR1	p.Thr4752=	chr19:38578001	rs1468571	Heterozygous	Tier2	Benign	0.1	—	0.0	VCV
SCN2A	c.4308+755A>G	chr2:165378405	rs116093037	Heterozygous	Tier2	Not reviewed	6.8	—	7.5	— /
SCN4A	p.His599Arg	chr17:63961242	rs187401185	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	24.4	0.925	27.8	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
SCN5A	p.Ile1761=	chr3:38551086	rs375323548	Heterozygous	Tier2	Conflicting classifications of pathogenicity	8.6	—	2.2	VCV
SDHA	c.1432+1198A>G	chr5:237797	rs112305500	Heterozygous	Tier2	Not reviewed	0.8	—	14.3	— /
SMARCA2	p.Gln228Pro	chr9:2039793	rs62534884	Heterozygous	Tier3	Benign	3.5	0.048	7.5	VCV
SMARCA4	c.4425-5369C>T	chr19:11052886	rs1529729	Heterozygous	Tier2	Not reviewed	0.6	—	1.7	— /
SMARCAL1	p.Ser315Arg	chr2:216420381	rs2066522	Heterozygous	Tier3	Benign	10.6	0.122	10.6	VCV
SMPX	c.*14+6246G>C	chrX:21731303	rs1015473	Homozygous	Tier2	Not reviewed	1.1	—	9.4	— /
SPEG	p.Val3081Met	chr2:219490812	rs375345784	Heterozygous	Tier2	Uncertain significance	25.2	0.387	23.8	VCV
SPTB	c.-377G>C	chr14:64880117	chr14:64880117	Heterozygous	Tier2	Not reviewed	19.1	—	13.7	— /
STAT1	c.1222-5T>C	chr2:190985665	rs191364028	Heterozygous	Tier2	Conflicting classifications of pathogenicity	13.9	—	16.9	VCV
SYNE2	p.Ile2564Val	chr14:64051603	rs11628107	Heterozygous	Tier3	Benign	2.4	0.067	6.1	VCV
SYNE2	p.Glu3026Asp	chr14:64052991	rs34843668	Heterozygous	Tier3	Benign	19.5	0.059	16.8	VCV
SYNE2	p.Pro4733Thr	chr14:64130105	rs75568433	Heterozygous	Tier3	Benign	23.3	0.155	21.2	VCV
SYNE2	p.Glu4913Lys	chr14:64137877	rs12881815	Heterozygous	Tier3	Benign	22.9	0.141	20.6	VCV
TANGO2	c.605+677G>C	chr22:20062360	rs951125062	Heterozygous	Tier2	Not reviewed	6.5	—	6.6	— /
TCF4	c.145+55622T>A	chr18:55529658	chr18:55529658	Heterozygous	Tier2	Not reviewed	19.2	—	15.5	— /
TCOF1	p.Lys1445del	chr5:150396822	rs574569798	Heterozygous	Tier3	Benign/Likely benign	—	—	—	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
TERT	c.2654+10G>A	chr5:1266454	rs375473823	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.1	—	0.1	VCV
TET2	p.Gly355Asp	chr4:105235006	rs61744960	Heterozygous	Tier3	Benign	23.0	0.138	20.1	VCV
TFR2	p.Arg752His	chr7:100621008	rs41295942	Heterozygous	Tier3	Benign/Likely benign	23.1	0.185	21.0	VCV
THSD4	p.Asp395Asn	chr15:71660560	rs72742760	Heterozygous	Tier2	Benign	24.7	0.237	20.8	VCV
TMEM126B	p.Ala198Val	chr11:85636129	rs17850847	Heterozygous	Tier3	Benign	0.0	0.087	3.4	VCV
TMEM231	p.Arg237Thr	chr16:75541410	rs199813223	Heterozygous	Tier2	Conflicting classifications of pathogenicity	23.3	0.403	27.0	VCV
TMEM270	p.Gly137Ser	chr7:73865329	rs73142829	Heterozygous	**Tier1**	Not reviewed	0.0	0.017	20.1	VCV
TMEM43	c.*115T>C	chr3:14141910	rs1043943	Heterozygous	Tier2	Benign/Likely benign	1.6	—	9.2	VCV
TMPO	p.Ala287Pro	chr12:98544517	rs7133258	Heterozygous	Tier3	Benign	22.5	0.067	22.3	VCV
TTN	p.Ser21716Leu	chr2:178584404	rs13021201	Heterozygous	Tier3	Benign/Likely benign	23.4	0.360	22.8	VCV
TTN	c.45083-10A>G	chr2:178621751	rs72677222	Heterozygous	**Tier1**	Benign/Likely benign	17.8	—	26.0	VCV
TTN	p.Glu5963Gly	chr2:178730645	rs146983095	Heterozygous	Tier2	Benign/Likely benign	22.8	0.110	20.6	VCV
TTN	p.Val2823Phe	chr2:178770234	rs33917087	Heterozygous	Tier2	Benign/Likely benign	23.0	0.424	22.9	VCV
TXNRD2	p.Gln37Leu	chr22:19931092	rs148092370	Heterozygous	Tier2	Benign/Likely benign	2.1	0.030	5.5	VCV

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Exte Acc
VCL	c.3153+47T>C	chr10:74114434	rs2270550	Heterozygous	Tier2	Benign	0.3	—	5.1	VCV
WAC	c.274+13301C>G	chr10:28549058	rs913821	Homozygous	Tier2	Not reviewed	0.1	—	0.5	— /
WFS1	p.Arg818Cys	chr4:6302247	rs35932623	Heterozygous	Tier2	Conflicting classifications of pathogenicity	27.1	0.758	24.3	VCV
WRN	p.Thr324Ala	chr8:31080997	rs1800390	Heterozygous	Tier2	Benign/Likely benign	0.0	0.018	0.1	VCV
WRN	c.2967+88G>A	chr8:31132594	rs2252480	Heterozygous	Tier2	Benign	1.3	—	0.7	VCV
YARS2	p.Ala35Asp	chr12:32755771	rs149447502	Heterozygous	Tier2	Benign/Likely benign	10.1	0.111	9.4	VCV
ZEB2	c.-69-197A>T	chr2:144517616	rs533595036	Heterozygous	Tier2	Not reviewed	20.7	—	18.6	— /
ZEB2	c.-69-198C>G	chr2:144517617	rs558617021	Heterozygous	Tier2	Not reviewed	17.8	—	15.7	— /
ZIC2	p.His238dup	chr13:99982754	chr13:99982754	Heterozygous	Tier3	Benign/Likely benign	—	—	—	VCV
ZNF462	c.*2246dup	chr9:107013270	chr9:107013270	Heterozygous	Tier2	Not reviewed	—	—	—	— /

4. Metabolic, Mitochondrial & DNA Repair Machinery

0 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	External Accessions

5. Protective & Longevity Modifiers

8 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)	REVEL	AVI (Phred)	Externa Accessi
ADH1C	p.Ile350Val	chr4:99339632	rs698	Heterozygous	Tier2	protective	15.4	—	15.9	VCV181
ADH1C	p.Arg272Gln	chr4:99342808	rs1693482	Heterozygous	Tier2	protective	10.4	—	12.8	VCV181
CASP8	c.-937_-932del	chr2:201232809	rs3834129	Heterozygous	Tier3	protective	—	—	—	VCV776
CCR5	c.-556A>G	chr3:46370444	rs1799987	Heterozygous	Tier2	Conflicting classifications of pathogenicity|protective	1.5	—	0.4	VCV818
CYP46A1	c.357-124G>A	chr14:99699891	rs3742377	Heterozygous	Tier3	protective	0.0	—	0.9	VCV169
LMO1	c.26-3357G>T	chr11:8233861	rs2168101	Heterozygous	Tier3	Benign|protective	18.7	—	22.9	VCV221
MPO	c.-643G>A	chr17:58281401	rs2333227	Homozygous	Tier3	protective|risk factor	2.5	—	2.5	VCV363
NOS3	c.-51-762C>T	chr7:150992991	rs2070744	Homozygous	Tier3	protective	0.6	—	7.8	VCV517

6. Secondary & Exploratory Clinical Variants

457 Variants

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
AAGAB	p.Met110Lys	chr15:67236440	rs186662479	Heterozygous	Tier2	Benign/Likely benign	22.6
ABCA2	p.Asp2047Asn	chr9:137010655	rs370257727	Heterozygous	Tier2	Uncertain significance	17.1
ABCB8	p.Ser39Asn	chr7:151033625	rs6975981	Heterozygous	Tier3	Benign	16.5
ABCC1	p.Ser667=	chr16:16079364	rs8187863	Heterozygous	Tier2	Benign	13.5
ABCC4	c.3871-6349C>T	chr13:95028031	rs4148546	Heterozygous	Tier2	Not reviewed	1.9
ACACB	p.Pro2194Leu	chr12:109260564	rs139082246	Heterozygous	**Tier1**	Not reviewed	25.5
ACVR1	c.67+4130T>A	chr2:157795297	rs1368063962	Heterozygous	Tier2	Not reviewed	5.7
ADAM8	p.Ser176ThrfsTer55	chr10:133273300	rs1283549753	Heterozygous	Tier2	Not reviewed	—
ADAMTS7	p.Leu1125Pro	chr15:78766537	rs201467377	Heterozygous	Tier2	Likely benign	0.0
ADAMTS7	p.Leu1125Met	chr15:78766538	rs199506661	Heterozygous	Tier2	Likely benign	14.4
ADAMTS7	p.Val1124Ala	chr15:78766540	rs200298138	Heterozygous	Tier2	Likely benign	0.0
ADGRF5	p.Pro416Ala	chr6:46872008	rs149458418	Heterozygous	Tier2	Not reviewed	0.0
AEBP1	p.Ala200Val	chr7:44107442	rs139352566	Heterozygous	Tier2	Benign/Likely benign	2.5
AEBP1	p.Val1148Ile	chr7:44114226	rs13898	Heterozygous	Tier3	Benign/Likely benign	20.9
AFDN	p.Ile438Val	chr6:167896967	rs35140809	Heterozygous	Tier2	Uncertain significance	25.5
AGPS	c.*3917_*3922del	chr2:177542102	rs369534713	Heterozygous	Tier2	Uncertain significance	—
AK4P3	p.Ala177Val	chr12:31615910	rs35037649	Heterozygous	Tier2	Not reviewed	7.1
AKR1A1	p.Ala69=	chr1:45566871	rs41269113	Heterozygous	Tier2	Not reviewed	8.7
ALAS1	p.Leu510Phe	chr3:52211480	rs762970207	Heterozygous	Tier2	Uncertain significance	23.5
ALDH4A1	c.*1197C>G	chr1:18871648	rs7419135	Heterozygous	Tier2	Uncertain significance	8.3

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
ALKBH3	p.Arg70Ter	chr11:43884007	rs145265812	Heterozygous	Tier2	Not reviewed	40.0
ALS2	p.Gly1247=	chr2:201718172	rs3219166	Heterozygous	Tier2	Conflicting classifications of pathogenicity	7.7
AMER1	p.His134Pro	chrX:64192886	rs146489129	Homozygous	Tier2	Benign/Likely benign	20.4
AMTN	p.Gly78Ser	chr4:70524899	rs151041998	Heterozygous	Tier2	Not reviewed	15.0
ANK2	c.85-22668G>A	chr4:113151748	rs11929834	Heterozygous	Tier2	Not reviewed	8.0
ANK2	p.Arg3906Trp	chr4:113373306	rs121912706	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	27.7
ANK3	p.Ile3867Val	chr10:60069282	rs761212770	Heterozygous	Tier2	Conflicting classifications of pathogenicity	1.2
ANTXR2	c.1429-3459A>G	chr4:79910926	rs78740643	Homozygous	Tier2	Not reviewed	5.6
ANTXR2	c.1428+33878T>A	chr4:79943743	rs77331798	Homozygous	Tier2	Not reviewed	3.0
ANTXR2	c.1041+10276A>G	chr4:79998245	rs11098983	Homozygous	Tier2	Not reviewed	2.3
ANTXR2	c.637-2754G>A	chr4:80038786	rs76275007	Homozygous	Tier2	Not reviewed	1.2
AP3B1	c.279+5115_279+5116del	chr5:78235746	rs1746849975	Heterozygous	Tier2	Not reviewed	—
APBA3	p.Ala79Val	chr19:3760029	rs11556574	Heterozygous	Tier2	Not reviewed	3.7
AR	p.Gln79_Gln80del	chrX:67545317	chrX:67545317	Homozygous	Tier3	Benign	—
ARL2BP	p.Glu87Lys	chr16:57249818	rs7198865	Heterozygous	Tier3	Benign/Likely benign	23.0
ASIC4	c.582+6977G>T	chr2:219522283	rs1422988144	Heterozygous	Tier2	Not reviewed	19.2
ASL	c.12+898G>A	chr7:66076991	rs146977222	Heterozygous	Tier3	Not reviewed	15.2
ATM	p.Ser49Cys	chr11:108227849	rs1800054	Heterozygous	Tier3	Benign	23.9
ATP11A	p.Arg585Gln	chr13:112842324	rs140975239	Heterozygous	Tier2	Not reviewed	23.3

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
ATP6V0A2	p.Asn850Ser	chr12:123758010	rs75279884	Heterozygous	Tier3	Benign	0.0
ATR	p.His90Tyr	chr3:142566145	rs28897763	Heterozygous	Tier2	Benign/Likely benign	21.2
ATXN1	c.-730+1246G>C	chr6:16760052	rs553861024	Heterozygous	Tier3	Not reviewed	19.5
AUH	c.262+2025C>T	chr9:91359603	rs117659478	Heterozygous	Tier2	Not reviewed	5.0
AXIN2	p.Asn412Ser	chr17:65537801	rs115931022	Heterozygous	Tier2	Benign/Likely benign	0.4
AXIN2	c.815+579G>C	chr17:65557227	chr17:65557227	Heterozygous	Tier2	Not reviewed	14.2
AZIN2	c.-73+113G>T	chr1:33081925	rs190309518	Heterozygous	**Tier1**	Not reviewed	0.4
B3GALT1	c.-409-38994A>G	chr2:167607915	rs1231445930	Heterozygous	Tier2	Not reviewed	17.5
B4GALT7	c.-77A>G	chr5:177600134	rs1407709637	Heterozygous	Tier2	Not reviewed	22.5
BCL11A	c.386-8965T>A	chr2:60477798	rs7593947	Homozygous	Tier2	Not reviewed	2.0
BEAN1	p.Leu31CysfsTer13	chr16:66469666	rs1300493594	Heterozygous	Tier2	Not reviewed	—
BEST3	p.Lys59AsnfsTer20	chr12:69694440	chr12:69694440	Heterozygous	Tier2	Not reviewed	—
BLM	p.Val724Ala	chr15:90765392	rs750687788	Heterozygous	**Tier1**	Uncertain significance	27.5
BLNK	p.Ala158Ser	chr10:96223879	rs148612299	Heterozygous	Tier2	Benign/Likely benign	0.1
BMP1	c.148+368_148+373dup	chr8:22165905	chr8:22165905	Heterozygous	Tier2	Not reviewed	—
BMPR1B	c.*2257C>G	chr4:95156930	rs551173634	Heterozygous	Tier2	Uncertain significance	0.4
BTN3A2	c.715+2T>G	chr6:26370605	rs58367598	Heterozygous	**Tier1**	Benign	28.7
C19orf12	p.Gly42Arg	chr19:29708290	rs200133991	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	24.3
C2CD3	p.Ser1663Asn	chr11:74057508	rs12419308	Heterozygous	Tier3	Benign	23.5
CA2	c.34+253G>A	chr8:85464368	rs1044764467	Heterozygous	Tier2	Not reviewed	11.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
CA8	c.418-13115A>G	chr8:60245494	rs9643367	Homozygous	Tier2	Not reviewed	16.8
CACNA1E	c.6027+27C>G	chr1:181793820	rs137964585	Heterozygous	Tier3	Benign	15.1
CACNA1H	p.Ala1966Val	chr16:1218979	rs72552054	Heterozygous	Tier3	Benign/Likely benign	9.9
CACNA1H	p.Ala2108Thr	chr16:1220254	rs56930660	Heterozygous	Tier2	Likely benign	18.7
CACNA2D1	c.294+36117T>C	chr7:82299018	rs6969265	Heterozygous	Tier2	Not reviewed	1.1
CACNA2D2	c.406-14564C>T	chr3:50408732	chr3:50408732	Heterozygous	Tier2	Not reviewed	13.3
CACNB1	p.Gly533Val	chr17:39175392	rs199798694	Heterozygous	Tier2	Not reviewed	21.7
CACNB2	p.Arg606Gly	chr10:18539557	rs61733968	Heterozygous	Tier3	Benign	21.4
CADM2	c.89-30901T>A	chr3:85771146	chr3:85771146	Heterozygous	Tier2	Not reviewed	14.6
CAMK2A	c.1238-1449C>G	chr5:150224666	rs13162545	Heterozygous	Tier2	Not reviewed	0.4
CAMK2D	c.1044+29T>C	chr4:113504947	rs1348055267	Heterozygous	Tier2	Not reviewed	20.7
CAMSAP1	p.Leu569Phe	chr9:135822956	rs138200991	Heterozygous	Tier2	Likely benign	6.6
CCDC90B	p.Val177Ile	chr11:83273804	rs140971996	Heterozygous	Tier2	Uncertain significance	21.6
CCR5	p.Ser185IlefsTer32	chr3:46373453	rs333	Heterozygous	Tier2	Benign	—
CD200	p.Ile86Thr	chr3:112345124	rs200570259	Heterozygous	Tier2	Not reviewed	22.3
CD226	p.Arg171His	chr18:69895916	rs79158855	Heterozygous	Tier2	Not reviewed	4.5
CD2AP	p.Leu331His	chr6:47579473	rs140188898	Heterozygous	Tier2	Conflicting classifications of pathogenicity	18.0
CDH1	c.1711+343C>T	chr16:68819768	rs4783689	Heterozygous	Tier2	Not reviewed	0.6
CDK10	p.Pro4Ala	chr16:89686720	rs187319559	Heterozygous	Tier2	Likely benign	7.4
CDK4	c.-60C>G	chr12:57752215	rs541752163	Heterozygous	Tier3	Benign	17.3

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
CEACAM5	p.Thr247Ile	chr19:41715686	chr19:41715686	Heterozygous	Tier2	Not reviewed	7.4
CELA1	p.Tyr5Ser	chr12:51346625	rs117443541	Heterozygous	Tier2	Not reviewed	14.8
CELA1	p.Tyr5His	chr12:51346626	rs116944010	Heterozygous	Tier2	Not reviewed	15.2
CELA1	p.Leu4AlafsTer21	chr12:51346630	rs377599213	Heterozygous	Tier2	Benign	0.8
CELA1	p.Val3ProfsTer21	chr12:51346631	rs1491233894	Heterozygous	Tier2	Benign	—
CELSR2	p.Arg2015Lys	chr1:109267578	rs72703203	Heterozygous	Tier3	Benign	18.3
CEP250	p.Glu881Lys	chr20:35490691	rs140439099	Heterozygous	Tier2	Benign/Likely benign	10.3
CEP250	p.Asp2006Asn	chr20:35504385	rs61729988	Heterozygous	Tier2	Benign/Likely benign	16.5
CERS3	p.Asp342Gly	chr15:100402840	rs1023783	Heterozygous	Tier3	Benign	3.6
CHD2	c.1153+109G>C	chr15:92944624	rs1320819	Homozygous	Tier2	Benign	0.2
CHD2	c.1502+505G>C	chr15:92949581	rs4327000	Homozygous	Tier2	Not reviewed	1.4
CHD3	p.Pro82dup	chr17:7885047	rs752425204	Heterozygous	Tier3	Benign	—
CHD3	c.*1202dup	chr17:7912779	chr17:7912779	Heterozygous	Tier2	Not reviewed	—
CHIT1	c.*119C>T	chr1:203216770	rs56391640	Heterozygous	Tier2	Uncertain significance	0.9
CHODL	c.-45+109054_-45+109055insGAA	chr21:18026454	chr21:18026454	Homozygous	Tier2	Not reviewed	—
CHRNB2	p.Gln397His	chr1:154572014	rs55685423	Heterozygous	Tier2	Conflicting classifications of pathogenicity	11.0
CLCN2	p.Arg558Ter	chr3:184354150	rs150362392	Heterozygous	Tier2	Not reviewed	38.0
CLYBL	c.62+31557A>G	chr13:99638314	rs1008004857	Heterozygous	Tier2	Not reviewed	11.0
CNGA3	p.Thr153Met	chr2:98389666	rs34314205	Heterozygous	Tier3	Benign/Likely benign	8.6
CNTN1	c.1380-149C>T	chr12:40943448	rs139915282	Heterozygous	**Tier1**	Likely benign	10.2

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
CNTN1	c.2420-1480G>A	chr12:41018857	rs146742800	Heterozygous	Tier2	Not reviewed	6.3
CNTN4	c.455-1339G>A	chr3:2865413	rs10510241	Heterozygous	Tier2	Not reviewed	6.7
CNTNAP3B	p.Cys1175Arg	chr9:41898919	rs1554733874	Homozygous	Tier2	Not reviewed	13.1
CNTNAP3B	p.Phe1173Leu	chr9:41898925	rs1554733876	Heterozygous	Tier2	Not reviewed	20.5
CNTNAP3B	p.Thr1051Ser	chr9:41908626	rs1554734101	Homozygous	Tier2	Not reviewed	10.7
COL22A1	p.Pro1589Thr	chr8:138589369	rs139658465	Heterozygous	Tier2	Not reviewed	20.2
COL23A1	c.675+2T>C	chr5:178262215	rs2973744	Heterozygous	**Tier1**	Not reviewed	33.0
COL4A2	p.Val1399Ile	chr13:110503903	rs45520539	Heterozygous	Tier3	Benign	0.1
COL4A2	c.4882-83T>C	chr13:110511851	rs2479426	Heterozygous	Tier3	Benign	1.5
COMP	p.Asn386Asp	chr19:18786630	rs61739916	Heterozygous	Tier3	Benign/Likely benign	25.5
COX20	p.Asn118Ser	chr1:244843172	rs61749963	Heterozygous	Tier3	Benign	3.3
CPS1	c.3+12249del	chr2:210489999	chr2:210489999	Heterozygous	Tier2	Not reviewed	—
CPS1	p.Pro1418Ser	chr2:210675818	rs150966847	Heterozygous	Tier2	Conflicting classifications of pathogenicity	17.4
CSF2RB	p.Pro603Thr	chr22:36937615	rs1801122	Heterozygous	Tier3	Benign	22.1
CTNNA1	c.1389+2076C>T	chr5:138906517	rs187795019	Heterozygous	Tier3	Likely benign	20.7
CUX1	p.Lys546Gln	chr7:102278021	rs118010189	Heterozygous	Tier2	Likely benign	26.8
CUX2	c.301+6815A>C	chr12:111270654	rs1265564	Heterozygous	Tier2	Not reviewed	0.7
CYP1B1	c.*1938G>A	chr2:38068784	rs570132783	Heterozygous	Tier2	Uncertain significance	0.2
CYP26C1	c.1191+605G>A	chr10:93066890	rs117929802	Heterozygous	Tier2	Not reviewed	4.9
CYP2E1	c.-118+195C>T	chr10:133520755	rs141625332	Heterozygous	**Tier1**	Not reviewed	2.5

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
CYP3A5	p.Ser100Tyr	chr7:99672599	rs41279857	Heterozygous	Tier2	Likely benign	23.2
CYP4V2	c.*1474G>T	chr4:186212115	rs77175303	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.4
DARS2	p.Gly338Glu	chr1:173839539	rs141298312	Heterozygous	Tier3	Benign	19.8
DCLRE1B	p.Arg283Cys	chr1:113911439	rs145569979	Heterozygous	Tier2	Likely benign	7.1
DCSTAMP	p.Arg439Trp	chr8:104355162	rs367714111	Heterozygous	Tier2	Not reviewed	17.3
DDR2	c.-1310T>C	chr1:162630329	rs7544972	Heterozygous	Tier2	Not reviewed	0.4
DDX59	c.1597-486_1597-484del	chr1:200625532	chr1:200625532	Heterozygous	Tier2	Not reviewed	—
DGKQ	c.2215-53A>C	chr4:962135	rs116777977	Homozygous	Tier2	Not reviewed	7.5
DIAPH1	p.Pro619_Pro620dup	chr5:141573997	chr5:141573997	Heterozygous	Tier3	Benign/Likely benign	—
DIAPH1	c.118-2199G>A	chr5:141590449	rs899162845	Heterozygous	Tier2	Not reviewed	5.9
DNASE2	p.His148Gln	chr19:12878737	chr19:12878737	Heterozygous	Tier2	Not reviewed	0.0
DPF3	c.33-32289_33-32288dup	chr14:72804185	chr14:72804185	Heterozygous	Tier2	Not reviewed	—
DPP4	c.1833-222_1833-219dup	chr2:162009520	chr2:162009520	Heterozygous	Tier2	Not reviewed	—
DUOX1	p.Arg194Gly	chr15:45135558	rs200124144	Heterozygous	Tier2	Likely benign	10.9
EDNRB	c.*1310C>T	chr13:77896890	rs3027095	Heterozygous	Tier2	Uncertain significance	0.6
EGFL7	c.572-141C>T	chr9:136670809	rs920684955	Heterozygous	**Tier1**	Not reviewed	0.6
EGFLAM	p.Arg444Gln	chr5:38409086	rs144190586	Heterozygous	Tier2	Not reviewed	17.9
EHD4	c.925-2404C>G	chr15:41912267	rs994695737	Heterozygous	Tier2	Not reviewed	14.9
ELOVL7	p.Ser79Tyr	chr5:60771922	rs368309905	Heterozygous	**Tier1**	Not reviewed	27.5
ELP1	p.Thr1292=	chr9:108874950	rs61749202	Heterozygous	Tier2	Conflicting classifications of pathogenicity	5.4

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
ELP1	p.Thr848Asn	chr9:108896997	rs10979599	Heterozygous	Tier2	Benign/Likely benign	17.9
ELP1	p.Arg405Trp	chr9:108911157	rs139703788	Heterozygous	Tier2	Conflicting classifications of pathogenicity	27.7
EMC1	c.96-708A>G	chr1:19245738	rs61764877	Homozygous	Tier2	Not reviewed	1.8
EMSY	c.877-501T>A	chr11:76472063	rs368463439	Heterozygous	Tier2	Not reviewed	13.0
EPCAM	c.492-5T>C	chr2:47377009	rs78608315	Heterozygous	Tier2	Benign/Likely benign	15.2
EPHA4	c.824-914G>A	chr2:221502086	rs1692506079	Heterozygous	Tier2	Not reviewed	16.1
ERRFI1	p.His21Arg	chr1:8015558	rs200259274	Heterozygous	Tier2	Not reviewed	16.8
ETV1	c.*606A>G	chr7:13895260	rs1020319686	Heterozygous	Tier2	Not reviewed	15.2
EYA1	c.34-25842C>T	chr8:71382353	rs10957550	Homozygous	Tier2	Not reviewed	8.8
EYA3	p.Ile504Val	chr1:27988565	chr1:27988565	Heterozygous	Tier2	Not reviewed	23.8
F10	c.502+774G>A	chr13:113141824	rs3213004	Heterozygous	Tier2	Not reviewed	1.6
F5	p.Thr295Ala	chr1:169556715	rs371760153	Heterozygous	**Tier1**	Uncertain significance	24.7
FASLG	c.*629_*632dup	chr1:172666617	chr1:172666617	Heterozygous	Tier2	Uncertain significance	—
FASN	p.Val1483Ile	chr17:82084916	rs2228305	Heterozygous	Tier3	Benign	0.0
FASTKD1	p.Ser768Ter	chr2:169531376	rs34291832	Heterozygous	Tier2	Benign	39.0
FBLIM1	p.Arg229Gln	chr1:15770553	rs200348478	Heterozygous	Tier2	Uncertain significance	22.6
FBN3	p.Arg2471His	chr19:8081044	rs3848570	Heterozygous	Tier3	Benign	22.2
FBXO41	p.Glu93Gly	chr2:73269353	rs1307947324	Heterozygous	Tier2	Not reviewed	27.4
FEZ1	c.-259+8188del	chr11:125584238	chr11:125584238	Heterozygous	Tier2	Not reviewed	—
FFAR3	p.Ser346Asn	chr19:35359927	rs201080710	Heterozygous	Tier2	Not reviewed	9.6

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
FGF9	c.*274_*277del	chr13:21701706	rs201279299	Heterozygous	Tier2	Uncertain significance	—
FHIT	c.-110-92900A>G	chr3:60914911	rs929917878	Heterozygous	Tier2	Not reviewed	20.5
FLG	p.Pro1169Leu	chr1:152311380	rs754087511	Heterozygous	Tier2	Not reviewed	8.1
FLG	p.Ser761CysfsTer36	chr1:152312601	rs558269137	Heterozygous	Tier2	Conflicting classifications of pathogenicity	—
FLT1	p.Glu982Ala	chr13:28322798	rs35832528	Heterozygous	Tier2	Not reviewed	22.9
FOXN1	p.His612Arg	chr17:28537324	rs200484835	Heterozygous	Tier2	Uncertain significance	22.1
FOXP2	c.169-15238dup	chr7:114519378	rs564368843	Heterozygous	Tier2	Not reviewed	—
GABRB3	c.240+5408G>A	chr15:26766994	rs3212335	Heterozygous	Tier2	Not reviewed	5.9
GANAB	p.Met594Val	chr11:62629642	rs76572368	Heterozygous	Tier3	Benign	0.7
GBP7	p.Met534Thr	chr1:89133319	rs374167387	Heterozygous	Tier2	Uncertain significance	12.8
GDAP1	c.-18+8599A>G	chr8:74330187	rs2975212	Homozygous	Tier2	Not reviewed	0.6
GDAP1	c.-18+12339A>G	chr8:74333927	rs2981271	Homozygous	Tier2	Not reviewed	0.3
GDAP1	c.-18+14480A>G	chr8:74336068	rs2975206	Homozygous	Tier2	Not reviewed	17.1
GDF1	p.Pro159dup	chr19:18869246	chr19:18869246	Heterozygous	Tier3	Benign	—
GFM2	p.Arg573His	chr5:74730268	rs143461990	Heterozygous	Tier2	Benign/Likely benign	17.8
GLIS3	c.1710+60058dup	chr9:4057711	rs35253627	Heterozygous	Tier2	Not reviewed	—
GLRB	c.122+6542A>G	chr4:157084688	rs72978482	Heterozygous	Tier2	Not reviewed	9.4
GM2A	p.Ala19Thr	chr5:151253271	rs1048719	Heterozygous	Tier3	Benign	9.7
GPR89B	p.Met89Thr	chr1:147943497	rs139258514	Heterozygous	Tier2	Uncertain significance	22.8
GRIA4	p.Gln28Arg	chr11:105611080	rs150577695	Heterozygous	**Tier1**	Not reviewed	23.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
GRIN2A	c.1122+16340T>C	chr16:9874646	rs7204266	Heterozygous	Tier2	Not reviewed	4.2
GRIN3B	p.Arg375Trp	chr19:1004624	rs753425600	Heterozygous	Tier2	Uncertain significance	25.0
GRM7	c.2698+22370A>T	chr3:7702665	rs776363586	Heterozygous	Tier2	Not reviewed	5.2
GTF3C3	p.Pro113dup	chr2:196793038	chr2:196793038	Heterozygous	Tier3	Likely benign	—
H6PD	p.Pro335Leu	chr1:9262317	rs140631516	Heterozygous	**Tier1**	Likely benign	25.7
HAPLN4	c.4-351A>G	chr19:19261914	rs141026619	Heterozygous	Tier2	Not reviewed	19.4
HBQ1	p.Pro115Ala	chr16:181022	rs144961211	Heterozygous	Tier2	Not reviewed	17.1
HEATR1	p.Ile1366Thr	chr1:236566857	rs115684254	Heterozygous	Tier2	Not reviewed	1.4
HEPHL1	p.Leu1037=	chr11:94110968	rs368314503	Heterozygous	Tier2	Not reviewed	6.6
HERC1	c.9224-9A>G	chr15:63659945	rs374571339	Heterozygous	Tier3	Benign	20.5
HEXA	p.Arg247Trp	chr15:72350584	rs121907970	Heterozygous	**Tier1**	Benign/Likely benign|other	25.1
HK3	c.915-177G>C	chr5:176889041	rs116035964	Heterozygous	Tier2	Not reviewed	5.5
HLCS	p.Arg471His	chr21:36936474	rs61732501	Heterozygous	Tier3	Benign	0.0
HTT	p.Gln35_Gln38del	chr4:3074877	chr4:3074877	Heterozygous	Tier3	Likely benign	—
HYCC1	c.*2904A>T	chr7:22942685	rs565649812	Heterozygous	Tier2	Uncertain significance	7.0
IGFBP5	p.Arg193His	chr2:216678221	rs755706987	Heterozygous	**Tier1**	Not reviewed	32.0
IL12RB2	c.76+12C>T	chr1:67320456	rs200381625	Heterozygous	Tier3	Likely benign	15.7
IL17RA	c.*4516C>T	chr22:17114336	rs180834857	Heterozygous	Tier2	Uncertain significance	0.5
IL17RA	c.*5778_*5779dup	chr22:17115588	chr22:17115588	Heterozygous	Tier2	Uncertain significance	—
IL1RAPL1	c.82+224347A>G	chrX:29013772	rs12843979	Homozygous	Tier2	Not reviewed	1.1

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
IL1RL2	p.Ala471Thr	chr2:102235010	rs75091099	Heterozygous	Tier2	Not reviewed	20.8
INPP5K	c.555-32A>G	chr17:1508258	rs74437970	Heterozygous	**Tier1**	Not reviewed	7.5
INVS	c.273+37788A>G	chr9:100164337	rs863668	Heterozygous	Tier2	Not reviewed	2.7
IQCA1	p.Cys693Arg	chr2:236338262	rs115960780	Heterozygous	**Tier1**	Not reviewed	26.2
ITGA6	c.-536+19132_-536+19137dup	chr2:172253377	chr2:172253377	Heterozygous	Tier2	Not reviewed	—
ITGAX	p.Pro383Leu	chr16:31362136	rs138501712	Heterozygous	Tier2	Not reviewed	19.4
ITGB6	p.Val438Met	chr2:160137782	rs61737764	Heterozygous	Tier2	Benign	25.0
ITPR3	c.89+6332C>T	chr6:33628023	chr6:33628023	Heterozygous	Tier2	Not reviewed	18.9
KANK2	p.Gly363Arg	chr19:11192993	rs763648935	Heterozygous	Tier2	Uncertain significance	23.3
KCNC2	c.687+20939G>A	chr12:75186358	rs181479386	Heterozygous	Tier2	Not reviewed	3.9
KCND2	c.1115+110760A>G	chr7:120386507	rs147760087	Heterozygous	Tier2	Not reviewed	8.8
KCND3	c.1107-86635A>G	chr1:111873741	rs2788612	Homozygous	Tier2	Not reviewed	18.2
KCND3	c.1106+86580A>G	chr1:111895041	rs10857913	Homozygous	Tier2	Not reviewed	0.5
KDM4D	p.Glu357Gly	chr11:94998442	rs76057256	Heterozygous	Tier2	Not reviewed	16.3
KNSTRN	p.Thr47Lys	chr15:40382975	rs199715714	Heterozygous	Tier2	Likely benign	0.7
KRT18	p.Ser230Thr	chr12:52951512	rs58472472	Heterozygous	Tier2	Likely benign	22.3
KY	p.Val186Ala	chr3:134620784	rs188233679	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.3
LAMA5	p.Lys812Thr	chr20:62335068	rs79319629	Heterozygous	Tier3	Benign	17.2
LARS2	p.Asp518Asn	chr3:45496303	rs116826217	Heterozygous	Tier2	Conflicting classifications of pathogenicity	28.8
LEF1	c.415-30738_415-30737del	chr4:108120000	chr4:108120000	Heterozygous	Tier2	Not reviewed	—

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
LHFPL5	c.*508A>T	chr6:35823473	rs371866936	Heterozygous	Tier2	Uncertain significance	0.7
LHX3	c.*780C>T	chr9:136196545	rs4842130	Heterozygous	Tier2	Uncertain significance	0.5
LIPT1	c.-14C>T	chr2:99155039	rs533919549	Heterozygous	Tier3	Likely benign	17.8
LNPEP	p.Leu369Gln	chr5:96986645	rs377307978	Heterozygous	Tier2	Uncertain significance	22.1
LPIN1	p.Pro646Ser	chr2:11802956	rs4669781	Heterozygous	Tier3	Benign	16.9
LRBA	c.1756-1308A>G	chr4:150901525	rs574372556	Heterozygous	Tier2	Not reviewed	4.4
LRP2	c.923-6G>A	chr2:169289151	rs144147038	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.6
LRRC26	p.Thr138Ala	chr9:137169532	rs149172628	Heterozygous	Tier2	Not reviewed	23.8
LTBP3	p.Ala438=	chr11:65552279	rs140740739	Heterozygous	Tier3	Likely benign	0.5
MACF1	p.Arg1239Leu	chr1:39317341	rs138873333	Heterozygous	Tier2	Benign/Likely benign	25.4
MACF1	p.His2191Tyr	chr1:39333159	rs138084833	Heterozygous	Tier2	Likely benign	10.4
MACF1	p.Ala5043Ser	chr1:39387969	rs144760259	Heterozygous	Tier2	Benign/Likely benign	24.1
MACF1	c.21959-173C>T	chr1:39479625	rs2275767	Heterozygous	Tier2	Benign	3.4
MACO1	c.*645T>C	chr1:25499111	rs1473415539	Heterozygous	Tier2	Not reviewed	13.5
MAGI2	c.1046-308C>T	chr7:78369521	rs7783086	Heterozygous	Tier2	Benign	0.3
MAP2	p.Glu55Lys	chr2:209653333	rs139310749	Heterozygous	Tier3	Benign/Likely benign	27.6
MAP2K6	c.16+11827A>C	chr17:69426827	rs997469618	Heterozygous	Tier2	Not reviewed	7.6
MARF1	p.Ser1620Phe	chr16:15598979	rs142983571	Heterozygous	Tier2	Not reviewed	28.0
MATR3	c.913-1304G>A	chr5:139313371	rs550294661	Heterozygous	Tier2	Not reviewed	16.0
MBD5	p.Gln1248Arg	chr2:148485940	rs143028540	Heterozygous	Tier2	Benign/Likely benign	26.2

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MBOAT2	c.222-2462A>C	chr2:8945726	rs10170080	Heterozygous	Tier2	Not reviewed	2.7
MCPH1	c.-1526del	chr8:6405127	chr8:6405127	Homozygous	Tier2	Not reviewed	—
MECOM	c.38-4338T>C	chr3:169385862	rs769582904	Heterozygous	Tier2	Not reviewed	18.4
MEF2B	p.Leu182=	chr19:19146871	rs3729770	Heterozygous	Tier2	Not reviewed	17.3
MEF2C	c.638-1909G>A	chr5:88733810	rs412458	Homozygous	Tier2	Not reviewed	0.3
MEF2C	c.-143+21974G>A	chr5:88860981	rs771966910	Heterozygous	Tier2	Not reviewed	15.8
MEGF10	p.Cys203=	chr5:127396728	rs113794264	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.7
MEIS2	c.754+3537C>A	chr15:37080234	rs1892019046	Heterozygous	Tier2	Not reviewed	17.1
MET	p.Asn375Ser	chr7:116700208	rs33917957	Heterozygous	Tier3	Benign/Likely benign	17.6
MGST3	p.Trp139Ser	chr1:165655461	rs199915214	Heterozygous	Tier2	Not reviewed	29.2
MLPH	c.445+2596T>C	chr2:237513697	rs75941539	Heterozygous	Tier2	Not reviewed	3.3
MMS22L	c.2040-12726A>G	chr6:97199416	chr6:97199416	Heterozygous	Tier2	Not reviewed	19.1
MNAT1	c.*1348A>G	chr14:60969697	rs2036747329	Heterozygous	Tier2	Not reviewed	10.2
MORF4L1	p.Arg74Gln	chr15:78886206	rs1203318217	Heterozygous	Tier2	Not reviewed	23.3
MRPL23	p.Arg113Cys	chr11:1962278	rs117979807	Heterozygous	Tier2	Not reviewed	1.3
MRPL23	p.Arg162Ser	chr11:1972745	rs190063	Homozygous	Tier2	Not reviewed	8.7
MSLN	p.Pro608Leu	chr16:768687	rs141160926	Heterozygous	Tier2	Likely benign	5.2
MTHFD1	p.Lys134Arg	chr14:64415662	rs1950902	Homozygous	Tier3	Benign	20.0
MTMR7	p.Met522Thr	chr8:17302209	rs117739013	Heterozygous	Tier2	Likely benign	21.7
MUC5B	p.Pro1472Leu	chr11:1241295	rs55669609	Heterozygous	Tier3	Benign	0.7

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MUC5B	p.Pro2890Ser	chr11:1245548	rs201287529	Heterozygous	Tier3	Benign	0.0
MUC5B	p.Thr3131Met	chr11:1246272	rs141653894	Heterozygous	Tier3	Benign	20.3
MYH10	c.663+3218G>A	chr17:8573425	rs7208821	Heterozygous	Tier2	Not reviewed	1.6
MYO5A	p.Arg1320Ser	chr15:52343197	rs61731219	Heterozygous	Tier3	Benign	23.2
MYOM1	p.Glu704Lys	chr18:3135646	rs149528866	Heterozygous	Tier2	Benign/Likely benign	25.2
MYOM1	c.1174+2180C>T	chr18:3171758	rs12965146	Heterozygous	Tier2	Not reviewed	3.4
MYOM1	p.Ser47Gly	chr18:3215085	rs202145133	Heterozygous	Tier2	Benign/Likely benign	23.4
MYRF	p.Arg920His	chr11:61781324	rs61747222	Heterozygous	Tier3	Benign	1.0
MYT1L	p.Glu167del	chr2:1942999	chr2:1942999	Heterozygous	Tier3	Benign	—
NAF1	p.Gly450Ser	chr4:163129034	rs200516616	Heterozygous	Tier3	Benign/Likely benign	6.7
NEDD4	p.Arg677fs	chr15:55839992	rs1463127951	Heterozygous	Tier2	Not reviewed	—
NFIA	c.560-5317T>C	chr1:61272203	rs530580492	Heterozygous	Tier2	Not reviewed	8.2
NFIA	c.626-26539G>A	chr1:61305973	rs2806434	Homozygous	Tier2	Not reviewed	1.8
NFIB	c.108+2797dup	chr9:14395729	chr9:14395729	Heterozygous	Tier2	Not reviewed	—
NFKBIB	p.Pro170Thr	chr19:38905424	rs751420371	Heterozygous	Tier2	Not reviewed	3.6
NKX6-2	p.Met22Leu	chr10:132785885	rs201425867	Heterozygous	Tier2	Benign/Likely benign	22.2
NLRP1	p.Val991=	chr17:5533976	rs11654427	Heterozygous	Tier2	Benign	7.3
NLRP11	p.Gln388Arg	chr19:55809447	rs111287552	Heterozygous	Tier2	Not reviewed	6.3
NOD2	p.Arg675Trp	chr16:50712015	rs2066844	Heterozygous	Tier2	Conflicting classifications of pathogenicity|association	23.6
NPAP1	p.Pro46Ser	chr15:24676003	rs199753715	Heterozygous	Tier2	Likely benign	15.3

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
NPC1L1	p.Thr499Met	chr7:44538901	rs35803101	Heterozygous	Tier2	Not reviewed	22.9
NPM1	c.58+514T>G	chr5:171388520	rs1770393339	Heterozygous	**Tier1**	Not reviewed	8.7
NQO1	p.Pro187Ser	chr16:69711242	rs1800566	Heterozygous	Tier2	Uncertain significance	24.2
NQO1	p.Arg139Trp	chr16:69714966	rs1131341	Heterozygous	Tier3	association	23.7
NR3C2	c.*378C>T	chr4:148080966	rs1033911893	Heterozygous	Tier2	Not reviewed	15.8
NRP1	c.2335-99T>C	chr10:33185823	rs12768811	Heterozygous	Tier2	Not reviewed	13.6
NRP1	c.248+3599C>A	chr10:33327109	rs2804497	Heterozygous	Tier2	Not reviewed	0.8
NRTN	c.-396_-388dup	chr19:5823767	rs576981082	Heterozygous	Tier2	Not reviewed	—
NRXN1	c.4128+39765G>A	chr2:50013506	rs3796036	Heterozygous	Tier2	Not reviewed	0.1
NRXN1	c.3365-51910A>G	chr2:50288880	rs1356888	Homozygous	Tier2	Not reviewed	1.8
NSMCE1	p.Cys231Tyr	chr16:27225755	rs774697446	Heterozygous	**Tier1**	Uncertain significance	25.6
NTNG1	c.1391-11644G>A	chr1:107468967	rs11185120	Heterozygous	Tier2	Not reviewed	0.8
NTSR1	p.Ala72Val	chr20:62709422	rs11698783	Heterozygous	Tier2	Benign	14.4
OBSL1	p.Arg368Cys	chr2:219568235	rs35009641	Heterozygous	Tier3	Benign/Likely benign	29.6
OCA2	c.2432+26353C>T	chr15:27818606	rs74409036	Heterozygous	Tier2	Not reviewed	5.5
OMD	p.Pro322Leu	chr9:92415453	chr9:92415453	Heterozygous	Tier2	Not reviewed	27.7
OPLAH	p.Gln931Arg	chr8:144053288	rs200702041	Heterozygous	Tier2	Likely benign	22.8
OR52E4	p.Met61Thr	chr11:5884474	rs755242702	Heterozygous	Tier2	Not reviewed	23.6
P3H2	c.824-1035T>A	chr3:189990073	rs55848820	Heterozygous	Tier2	Not reviewed	9.0
PAK3	p.Thr247Pro	chrX:111163700	chrX:111163700	Heterozygous	**Tier1**	Not reviewed	25.1
PALLD	p.Arg303fs	chr4:168599976	rs1170255696	Heterozygous	Tier2	Not reviewed	—

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PARP3	p.Arg140Gln	chr3:51944496	rs761611493	Heterozygous	**Tier1**	Not reviewed	24.8
PAX7	c.1156-13208G>C	chr1:18722424	rs2089505906	Heterozygous	Tier2	Not reviewed	20.3
PCDHA1	c.2394+93408A>T	chr5:140882092	rs59479	Heterozygous	Tier2	Not reviewed	20.9
PCDHGA9	p.Pro256Ser	chr5:141403718	rs77227638	Heterozygous	Tier2	Not reviewed	13.0
PCK1	p.Val368Ile	chr20:57564309	rs1804160	Heterozygous	Tier3	Benign	7.2
PGM2L1	c.1633-396T>A	chr11:74338997	rs75285093	Heterozygous	Tier2	Not reviewed	15.1
PHIP	c.3379+491A>G	chr6:78965212	rs10943608	Homozygous	Tier2	Not reviewed	1.3
PIEZO1	p.Arg1211Cys	chr16:88726783	rs142506765	Heterozygous	Tier2	Conflicting classifications of pathogenicity	23.2
PIEZO1	p.Glu756dup	chr16:88733965	chr16:88733965	Heterozygous	Tier2	Conflicting classifications of pathogenicity	—
PIP5KL1	p.Thr304Met	chr9:127925113	rs149767177	Heterozygous	Tier2	Not reviewed	1.9
PJA2	p.Ala402Thr	chr5:109378283	rs34621137	Heterozygous	Tier2	Not reviewed	22.5
PKD1L3	c.1271+23G>T	chr16:71979984	rs72803747	Heterozygous	**Tier1**	Not reviewed	3.4
PKD2	p.Phe482Cys	chr4:88046767	rs75762896	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.9
PKP3	p.Arg185Trp	chr11:397054	rs151265573	Heterozygous	Tier2	Likely benign	21.2
PLA2G6	c.2034+355G>A	chr22:38115172	rs147795054	Heterozygous	**Tier1**	Not reviewed	1.7
PLB1	c.-114+4401G>A	chr2:28461598	rs56116810	Heterozygous	**Tier1**	Not reviewed	3.5
PLCD1	p.Asp231Asn	chr3:38011313	rs61755441	Heterozygous	Tier3	Benign	19.9
PLCXD3	p.Ser245Cys	chr5:41381904	rs1210486304	Heterozygous	**Tier1**	Uncertain significance	27.6
PLD1	p.Arg398Cys	chr3:171699780	rs143845082	Heterozygous	Tier2	Benign	32.0

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PLEKHG2	p.Arg647Cys	chr19:39422993	rs148805217	Heterozygous	Tier2	Uncertain significance	13.6
PLEKHO1	p.Arg279Gln	chr1:150159129	rs41264971	Heterozygous	Tier2	Not reviewed	23.7
PML	p.Lys725Arg	chr15:74044533	rs111398700	Heterozygous	Tier2	Likely benign	18.2
POLE	p.Gly6Arg	chr12:132687300	rs202220778	Heterozygous	Tier2	Conflicting classifications of pathogenicity	9.1
POR	c.*240C>T	chr7:75986721	rs13921	Heterozygous	Tier2	Uncertain significance	4.7
POU4F1	p.Leu46Phe	chr13:78602539	rs142455133	Heterozygous	Tier2	Likely benign	24.3
POU6F2	c.-435A>G	chr7:38977519	chr7:38977519	Heterozygous	Tier2	Not reviewed	18.8
PRDM5	c.93+202C>T	chr4:120922314	rs3761726	Heterozygous	Tier2	Likely benign	15.4
PRELP	p.Arg206Lys	chr1:203483801	rs41304277	Heterozygous	Tier2	Not reviewed	0.1
PRKDC	p.Arg2899Cys	chr8:47826744	rs4278157	Heterozygous	Tier3	Benign	23.7
PRLR	p.Ile170Leu	chr5:35072610	rs72478580	Heterozygous	Tier2	Conflicting classifications of pathogenicity	7.7
PROX1	c.1833+1174A>C	chr1:214006446	rs1323155233	Heterozygous	Tier2	Not reviewed	16.4
PROZ	p.Arg295His	chr13:113171786	rs3024772	Heterozygous	Tier3	Likely benign	13.4
PRR12	p.Pro1516Thr	chr19:49601691	rs142810131	Heterozygous	Tier3	Benign	6.7
PRRX1	p.Val175Met	chr1:170726325	rs201365132	Heterozygous	Tier2	Uncertain significance	27.7
PTDSS2	c.367+914G>A	chr11:474891	rs142613557	Heterozygous	Tier2	Not reviewed	0.9
PTDSS2	p.Glu232Gly	chr11:488272	rs908582716	Heterozygous	**Tier1**	Not reviewed	31.0
PTGER3	p.Asn395LysfsTer9	chr1:70952979	rs568967213	Heterozygous	Tier2	Likely benign	—
PTGFR	c.-72-20273C>T	chr1:78472399	rs898436651	Heterozygous	Tier2	Not reviewed	6.6
PTK7	p.Glu745Asp	chr6:43143604	rs9472017	Heterozygous	Tier3	Benign	26.2

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PTPN22	p.Gln456Glu	chr1:113838034	rs72650672	Heterozygous	Tier2	Likely benign	22.4
PTX3	p.Gln167Arg	chr3:157437882	rs191352729	Heterozygous	Tier3	Benign	3.8
RAB5IF	c.349-875G>A	chr20:36611135	rs116979051	Heterozygous	Tier2	Not reviewed	0.5
RAD51AP1	c.871+1G>C	chr12:4556503	rs371887719	Heterozygous	Tier2	Not reviewed	33.0
RB1CC1	p.Asn1314Lys	chr8:52645747	rs34701924	Heterozygous	Tier2	Benign/Likely benign	21.6
RELN	p.Ser630Arg	chr7:103651665	rs115734214	Heterozygous	Tier3	Benign	25.1
REV3L	p.Asp1812His	chr6:111372921	rs3218599	Heterozygous	Tier3	Benign	18.5
RFX2	p.Gly156Val	chr19:6040035	rs143838272	Heterozygous	Tier2	Likely benign	24.3
RFX7	p.Leu1353Pro	chr15:56093670	rs33984059	Heterozygous	Tier3	Not reviewed	27.0
RILPL2	c.606-1G>A	chr12:123410650	rs28664930	Heterozygous	Tier2	Not reviewed	1.3
RIMS2	p.Asp1081Ala	chr8:103989394	chr8:103989394	Heterozygous	Tier2	Not reviewed	23.8
RIPOR2	p.Trp15Arg	chr6:25041884	rs148823241	Heterozygous	Tier3	Benign/Likely benign	29.4
RNF13	p.Val173Ile	chr3:149911994	rs150196894	Heterozygous	Tier3	Benign	18.8
RNF213	p.Thr1705Lys	chr17:80339481	rs147868237	Heterozygous	Tier2	Benign/Likely benign	23.5
RNF213	p.Gln2184Arg	chr17:80344886	rs138595111	Heterozygous	Tier2	Benign/Likely benign	1.0
RORA	p.Thr56fs	chr15:61059983	rs1424804388	Heterozygous	Tier2	Not reviewed	—
SACS	p.Ala303=	chr13:23355703	rs41315020	Heterozygous	Tier2	Conflicting classifications of pathogenicity	2.9
SAR1B	c.244+36_244+49del	chr5:134612642	chr5:134612642	Heterozygous	Tier2	Uncertain significance	—
SASH1	c.862+30dup	chr6:148514460	chr6:148514460	Heterozygous	Tier2	Not reviewed	—
SCN2B	c.71-2274A>G	chr11:118171025	rs149794440	Heterozygous	Tier2	Not reviewed	1.7

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SCNN1A	p.Arg181Trp	chr12:6363586	rs55797039	Homozygous	Tier3	Benign/Likely benign	18.6
SEC24D	p.Ala218Gly	chr4:118815471	rs35392900	Heterozygous	Tier3	Benign/Likely benign	16.8
SENP7	c.796+398A>G	chr3:101371610	rs918386520	Heterozygous	Tier2	Not reviewed	17.3
SEPTIN9	c.76+23238A>C	chr17:77330435	rs892964	Heterozygous	Tier2	Not reviewed	2.3
SERPINC1	p.Gln337=	chr1:173909694	rs5878	Heterozygous	Tier2	Benign	9.2
SETD9	c.98+5G>A	chr5:56909748	rs141233841	Heterozygous	Tier2	Not reviewed	23.4
SH3TC2	c.*1616C>T	chr5:149003095	rs758753462	Heterozygous	Tier2	Uncertain significance	4.7
SIK1	p.Gly15Ser	chr21:43426136	rs3746951	Heterozygous	Tier2	Benign	14.3
SLC15A4	c.1573+202G>A	chr12:128799057	rs572449834	Heterozygous	Tier2	Not reviewed	1.4
SLC15A4	c.1573+186G>T	chr12:128799073	rs541089091	Heterozygous	Tier2	Not reviewed	0.1
SLC17A2	p.Met452Leu	chr6:25913400	rs150509035	Heterozygous	Tier2	Not reviewed	13.7
SLC22A23	c.1083-1786_1083-1781del	chr6:3300000	chr6:3300000	Heterozygous	Tier2	Not reviewed	—
SLC22A23	c.654+13013A>G	chr6:3442893	rs1771687032	Heterozygous	Tier2	Not reviewed	13.0
SLC25A12	p.Arg141Gln	chr2:171844412	rs758739467	Heterozygous	Tier2	Uncertain significance	25.8
SLC26A4	c.*1836C>T	chr7:107717282	rs565613619	Heterozygous	Tier2	Uncertain significance	2.2
SLC26A9	p.Asn501Ser	chr1:205923608	rs148928330	Heterozygous	Tier2	Not reviewed	23.2
SLC27A3	p.Arg471Trp	chr1:153778517	rs137908170	Heterozygous	Tier2	Not reviewed	26.2
SLC4A4	c.-2+1203C>T	chr4:71093995	rs1742463061	Heterozygous	Tier2	Not reviewed	18.8
SLC5A11	p.Phe258Leu	chr16:24890978	rs35993597	Heterozygous	Tier3	Not reviewed	20.8
SLC6A4	c.1818+1348A>G	chr17:30201824	rs11080121	Heterozygous	Tier2	Not reviewed	5.6

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
SLC7A9	p.Ala324=	chr19:32858445	rs61730903	Heterozygous	Tier2	Conflicting classifications of pathogenicity	0.7
SLC9A2	p.Arg539Trp	chr2:102701098	rs371605244	Heterozygous	**Tier1**	Not reviewed	26.3
SLFN11	p.Leu564His	chr17:35353567	rs201829415	Heterozygous	Tier2	Not reviewed	20.3
SNCA	c.306+22657C>T	chr4:89799589	rs1191084189	Heterozygous	Tier2	Not reviewed	3.3
SORL1	p.Asp140Asn	chr11:121478133	rs140888526	Heterozygous	Tier2	Uncertain significance	24.6
SP110	c.1349-1241T>C	chr2:230179496	rs967006	Homozygous	Tier2	Not reviewed	3.4
SPG11	p.Lys1013Glu	chr15:44615364	rs111347025	Heterozygous	Tier2	Conflicting classifications of pathogenicity	23.0
SPTLC1	c.1136+848G>A	chr9:92045151	rs764656955	Heterozygous	Tier2	Not reviewed	14.5
ST8SIA1	p.Gln59Arg	chr12:22334057	rs780055957	Heterozygous	Tier2	Not reviewed	22.4
SUCLA2	c.*214+32057T>C	chr13:47873272	rs1575640	Heterozygous	Tier2	Not reviewed	3.3
SUCLA2	c.*214+8995G>A	chr13:47896334	rs7334463	Heterozygous	Tier2	Not reviewed	0.0
SUCLA2	c.*214+3557T>C	chr13:47901772	rs7337764	Heterozygous	Tier2	Not reviewed	8.0
SUCLA2	c.*27-6345A>G	chr13:47913480	rs9526410	Heterozygous	Tier2	Not reviewed	1.0
SUCLA2	c.*27-16018C>T	chr13:47923153	rs9591123	Heterozygous	Tier2	Not reviewed	1.2
SV2A	c.-95G>A	chr1:149913935	rs41264628	Heterozygous	Tier2	Not reviewed	4.7
SYT1	c.*825_*834dup	chr12:79449926	chr12:79449926	Heterozygous	Tier2	Not reviewed	—
TBL1XR1	c.*2309T>C	chr3:177023189	rs543404725	Heterozygous	Tier2	Not reviewed	16.3
TBP	p.Gln95del	chr6:170561959	chr6:170561959	Homozygous	Tier3	Benign	—
TBXAS1	p.Glu449Lys	chr7:140015841	rs8192868	Heterozygous	Tier3	Benign	23.9
TCAIM	p.Arg217His	chr3:44392932	rs142202431	Heterozygous	Tier2	Uncertain significance	28.6

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
TCF7L2	c.875+2274C>G	chr10:113148371	rs1409481450	Heterozygous	Tier2	Not reviewed	19.4
TECRL	p.Pro45Ser	chr4:64409219	rs61744866	Heterozygous	Tier3	Benign/Likely benign	14.9
TEK	p.Val600Leu	chr9:27197488	rs35030851	Heterozygous	Tier3	Benign	17.9
TENM2	c.-490_-476dup	chr5:166979201	chr5:166979201	Heterozygous	Tier2	Not reviewed	—
TENM2	p.Arg1266Gln	chr5:168204594	rs141103827	Heterozygous	Tier2	Not reviewed	24.9
TENM4	p.Pro935Arg	chr11:78738523	rs141706152	Heterozygous	**Tier1**	Conflicting classifications of pathogenicity	24.0
TENM4	p.Pro485Ser	chr11:78855981	rs191549326	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.3
TENT4A	p.Leu683=	chr5:6752900	rs28381429	Heterozygous	Tier2	Not reviewed	8.4
TET1	p.Leu73Phe	chr10:68572555	rs138684329	Heterozygous	Tier2	Not reviewed	20.0
TFEB	c.-22-1690C>T	chr6:41692925	rs769095757	Heterozygous	Tier2	Not reviewed	17.9
THBS3	c.1099-40C>T	chr1:155202074	rs182720155	Heterozygous	Tier2	Not reviewed	14.2
TIRAP	p.Ser180Leu	chr11:126292948	rs8177374	Heterozygous	Tier3	Benign	13.5
TMC2	p.Trp217Arg	chr20:2579149	rs34884202	Heterozygous	Tier3	Not reviewed	24.7
TNPO3	p.His693=	chr7:128975918	rs199826657	Heterozygous	Tier2	Conflicting classifications of pathogenicity	3.7
TNR	c.499+594_499+597del	chr1:175405623	chr1:175405623	Heterozygous	Tier2	Not reviewed	—
TNRC18	p.Ala300Val	chr7:5388925	rs149434477	Heterozygous	Tier2	Likely benign	6.5
TNXB	p.Ala1521Val	chr6:32073766	rs146120469	Homozygous	Tier2	Benign/Likely benign	7.0
TOM1	c.1284+86A>G	chr22:35345870	rs133399	Heterozygous	Tier3	Benign	2.7
TPM4	c.133-1860T>G	chr19:16080053	rs16981095	Heterozygous	Tier2	Not reviewed	3.0

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
TRAK1	c.92-10068T>A	chr3:42115352	rs17220321	Heterozygous	**Tier1**	Not reviewed	17.7
TRAP1	p.Asp685Asn	chr16:3658191	rs61756352	Heterozygous	Tier2	Conflicting classifications of pathogenicity	24.6
TREML2	p.Gly201Ter	chr6:41194609	chr6:41194609	Heterozygous	Tier2	Not reviewed	35.0
TRMT10A	p.Arg69His	chr4:99558191	rs147815779	Heterozygous	Tier2	Benign/Likely benign	21.7
TRPC7	c.781-702T>A	chr5:136316481	chr5:136316481	Heterozygous	Tier2	Not reviewed	17.7
TRPC7	c.-434_-433dup	chr5:136365689	chr5:136365689	Heterozygous	Tier2	Not reviewed	—
TRPM1	p.Leu9Phe	chr15:31160935	rs118047113	Heterozygous	Tier3	Benign	15.3
TRPV4	p.Pro19Ser	chr12:109814742	rs3742030	Heterozygous	Tier3	Benign	15.9
TSEN54	p.Ser302=	chr17:75521987	rs774219646	Heterozygous	Tier2	Not reviewed	12.6
TTC39B	c.77+18701A>T	chr9:15249213	rs147473128	Heterozygous	Tier3	Not reviewed	20.1
UBE2A	c.-157-36745C>T	chrX:119536898	rs1967298	Homozygous	Tier2	Not reviewed	1.5
UCK2	c.99+808G>T	chr1:165828740	rs562989479	Heterozygous	**Tier1**	Not reviewed	6.9
UNC80	p.Ala3227Val	chr2:209994236	rs368594548	Heterozygous	Tier2	Uncertain significance	25.9
UNC93B1	p.Arg519Leu	chr11:67991784	rs1451840915	Heterozygous	Tier2	Not reviewed	14.7
UQCC2	c.138+87G>T	chr6:33711462	rs7759182	Heterozygous	Tier2	Benign	15.7
VAPB	c.*5250T>C	chr20:58449485	rs6015274	Heterozygous	Tier2	Benign	9.5
VLDLR	c.-171G>C	chr9:2622019	rs35763266	Heterozygous	Tier3	Likely benign	16.4
VPS13B	c.5221-8292C>A	chr8:99633519	rs1828934932	Heterozygous	Tier2	Not reviewed	16.8
VPS13B	p.Arg1740fs	chr8:99639998	chr8:99639998	Heterozygous	Tier2	Not reviewed	—
VPS13C	p.Ile3726Val	chr15:61854543	rs115819951	Heterozygous	Tier2	Benign/Likely benign	23.6

Gene	Variant	Coordinate	rsID	Zygosity	Tier	ClinVar	CADD (Phred)
VPS13C	p.Arg2482His	chr15:61920099	rs115481870	Heterozygous	Tier2	Benign	25.6
VPS13C	p.Asn1955His	chr15:61934224	rs112236709	Heterozygous	Tier3	Benign	16.5
VPS18	p.Arg914Trp	chr15:40903159	rs189855795	Heterozygous	Tier2	Not reviewed	22.4
VPS33B	c.1226-136A>C	chr15:91003267	rs4244883	Homozygous	Tier2	Benign	5.3
VWA8	p.Arg668Gln	chr13:41811285	rs138075452	Heterozygous	Tier2	Conflicting classifications of pathogenicity	26.4
VWF	c.658-1399A>G	chr12:6076950	rs7135976	Heterozygous	Tier2	Not reviewed	4.7
WDR1	p.His248Arg	chr4:10087915	rs41268387	Heterozygous	Tier2	Benign	19.0
WDR19	p.Phe787=	chr4:39234873	rs200133722	Heterozygous	Tier2	Conflicting classifications of pathogenicity	14.5
WDR47	p.Ser498Leu	chr1:108995778	rs141359680	Heterozygous	Tier2	Not reviewed	19.7
WDR48	c.1475-1155C>T	chr3:39086973	chr3:39086973	Heterozygous	Tier2	Not reviewed	4.8
WWTR1	c.-179A>G	chr3:149657940	rs1050866729	Heterozygous	Tier2	Not reviewed	22.1
ZAN	p.His1922Tyr	chr7:100773850	rs314299	Heterozygous	Tier2	Benign	18.2
ZDHHC5	p.Pro45Ala	chr11:57682450	rs150741412	Heterozygous	Tier2	Uncertain significance	22.6
ZFHX3	c.-1709_-1696dup	chr16:73891745	chr16:73891745	Heterozygous	Tier2	Not reviewed	—
ZFYVE26	p.Arg2411His	chr14:67752483	rs34373049	Heterozygous	Tier3	Benign/Likely benign	23.8
ZNF205	p.Glu64Val	chr16:3115488	rs367655350	Heterozygous	Tier2	Uncertain significance	19.4
ZNF292	p.Val2435Ile	chr6:87260932	rs61744959	Heterozygous	Tier2	Not reviewed	5.6
ZNF469	p.Lys2559Glu	chr16:88435145	rs146789160	Heterozygous	Tier2	Benign/Likely benign	0.1
ZNF469	c.*634C>A	chr16:88439966	rs74032868	Heterozygous	Tier2	Uncertain significance	3.7